

Heapsort

Übungsblatt mit steigender Schwierigkeit
(Automatisch generiert)

Algorithmen und Datenstrukturen I

Hinweise

Heapsort-Algorithmus (Max-Heap):

1. **Anfangsheap herstellen:** Von $i = n/2$ bis 0: `siftdown(a, i, n)`
2. **Sortieren:** Für $i = n - 1$ bis 1:
 - Tausche `a[0]` (Maximum) mit `a[i]`
 - `siftdown(a, 0, i)`

Siftdown-Operation:

- Vergleiche Element mit seinen Kindern
- Wenn ein Kind grösser ist, tausche mit dem grösseren Kind
- Wiederhole bis Heapbedingung erfüllt oder Blatt erreicht

Heap-Bedingung: $a[i] \geq a[2i + 1]$ und $a[i] \geq a[2i + 2]$

Farbmarkierungen:

- Gelb = Element im Heap
- Grün = Sortierter Bereich (Endposition)
- Rot = Element das versickert
- Orange = Elemente die getauscht werden

Aufgabe 1 – Schwierigkeit: ★★

Klausuraufgabe WS14/15

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

19, 13, 14, 7, 8, 12, 3, 2

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

Fragen:

1. Geben Sie den Anfangsheap (Heap8) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap7, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgefuehrt?

Loesung:

Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

=== Phase 1: Anfangsheap herstellen ===

Schritt 1: siftdown(a, 4, 8): Versickere a[4]=8

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 2: a[4]=8 ist Blatt (keine Kinder), fertig

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 3: siftdown(a, 3, 8): Versickere a[3]=7

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 4: Nur linkes Kind vorhanden: a[7]=2

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 5: a[3]=7 >= a[7]=2 (groesseres Kind), Heap-Bedingung erfuehlt

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 6: siftdown(a, 2, 8): Versickere $a[2]=14$

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 7: Kinder: $a[5]=12$ und $a[6]=3$. Linkes Kind $12 \geq$ rechtes Kind 3, wachle linkes Kind

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 8: $a[2]=14 \geq a[5]=12$ (groesseres Kind), Heap-Bedingung erfuehlt

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 9: siftdown(a, 1, 8): Versickere $a[1]=13$

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 10: Kinder: $a[3]=7$ und $a[4]=8$. Rechtes Kind $8 >$ linkes Kind 7, wachle rechtes Kind

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 11: $a[1]=13 \geq a[4]=8$ (groesseres Kind), Heap-Bedingung erfuehlt

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 12: siftdown(a, 0, 8): Versickere $a[0]=19$

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

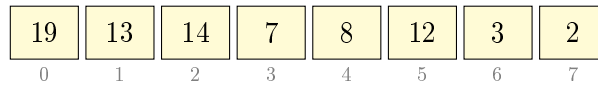
Schritt 13: Kinder: $a[1]=13$ und $a[2]=14$. Rechtes Kind $14 >$ linkes Kind 13, wachle rechtes Kind

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

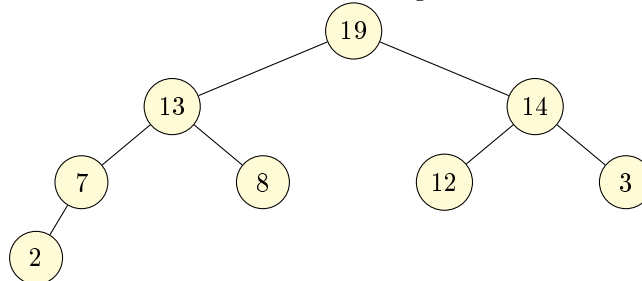
Schritt 14: $a[0]=19 \geq a[2]=14$ (groesseres Kind), Heap-Bedingung erfuehlt

19	13	14	7	8	12	3	2
0	1	2	3	4	5	6	7

Schritt 15: Heap8: Anfangsheap hergestellt

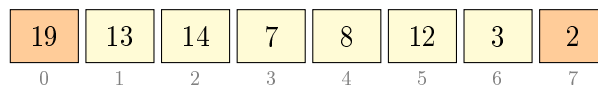


Baumdarstellung:

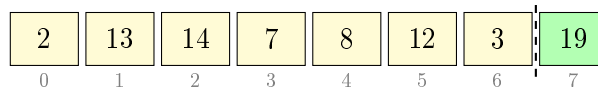


=== Phase 2: Sortieren durch Auswahl ===

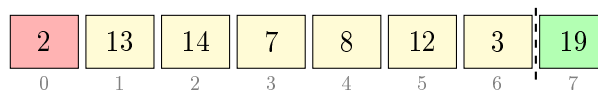
Schritt 16: Tausche Maximum $a[0]=19$ mit $a[7]=2$



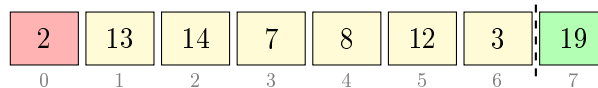
Schritt 17: $a[7]=19$ ist an Endposition, Heapgrösse: 7



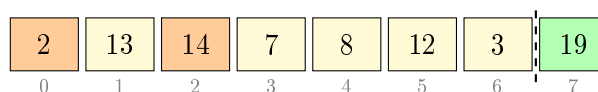
Schritt 18: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=2$



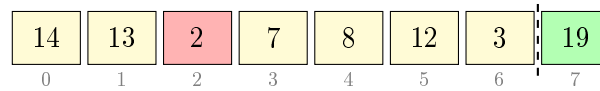
Schritt 19: Kinder: $a[1]=13$ und $a[2]=14$. Rechtes Kind $14 >$ linkes Kind 13 , wähle rechtes Kind



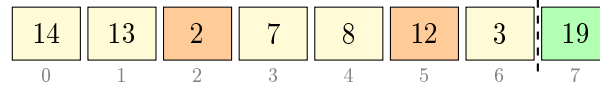
Schritt 20: $a[0]=2 < a[2]=14$ (grösseres Kind), tausche



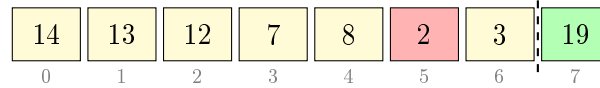
Schritt 21: Kinder: $a[5]=12$ und $a[6]=3$. Linkes Kind $12 \geq$ rechtes Kind 3 , wähle linkes Kind



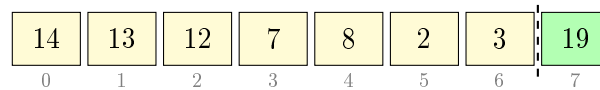
Schritt 22: $a[2]=2 < a[5]=12$ (groesseres Kind), tausche



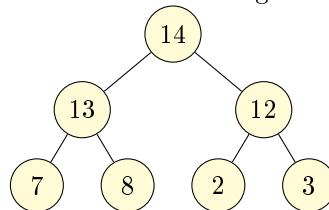
Schritt 23: $a[5]=2$ ist Blatt (keine Kinder), fertig



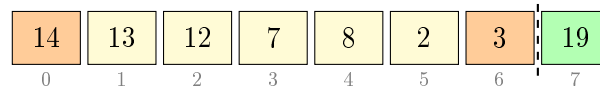
Schritt 24: Heap7: [14, 13, 12, 7, 8, 2, 3] | sortiert: [19]



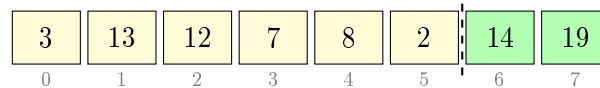
Baumdarstellung:



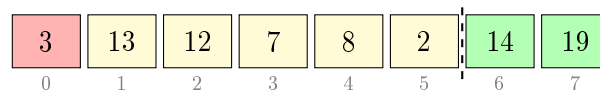
Schritt 25: Tausche Maximum $a[0]=14$ mit $a[6]=3$



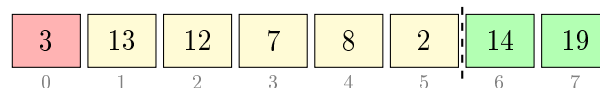
Schritt 26: $a[6]=14$ ist an Endposition, Heapgroesse: 6



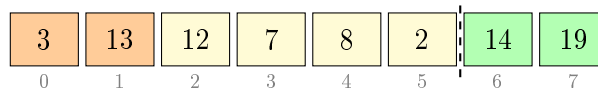
Schritt 27: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=3$



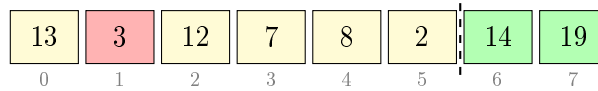
Schritt 28: Kinder: $a[1]=13$ und $a[2]=12$. Linkes Kind $13 \geq$ rechtes Kind 12, waehle linkes Kind



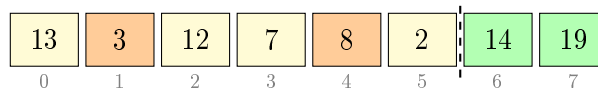
Schritt 29: $a[0]=3 < a[1]=13$ (groesseres Kind), tausche



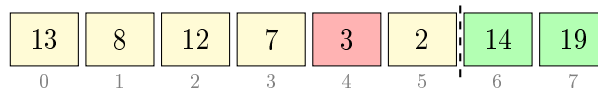
Schritt 30: Kinder: $a[3]=7$ und $a[4]=8$. Rechtes Kind $8 >$ linkes Kind 7, wachle rechtes Kind



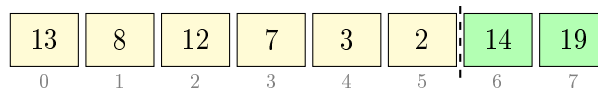
Schritt 31: $a[1]=3 < a[4]=8$ (groesseres Kind), tausche



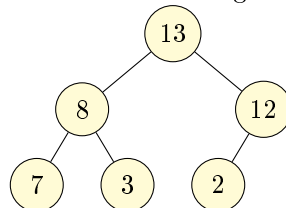
Schritt 32: $a[4]=3$ ist Blatt (keine Kinder), fertig



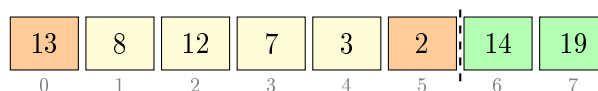
Schritt 33: Heap6: [13, 8, 12, 7, 3, 2] | sortiert: [14, 19]



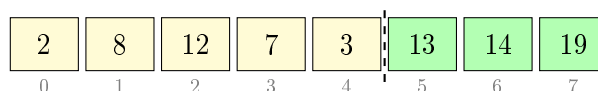
Baumdarstellung:



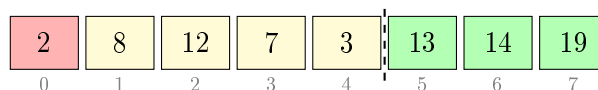
Schritt 34: Tausche Maximum $a[0]=13$ mit $a[5]=2$



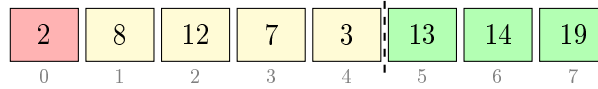
Schritt 35: $a[5]=13$ ist an Endposition, Heapgroesse: 5



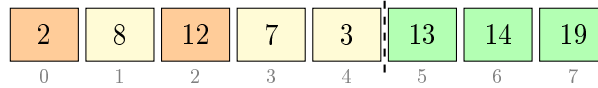
Schritt 36: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=2$



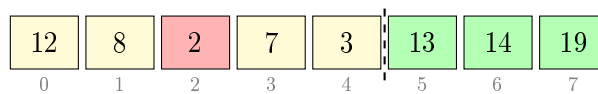
Schritt 37: Kinder: $a[1]=8$ und $a[2]=12$. Rechtes Kind $12 >$ linkes Kind 8, waehle rechtes Kind



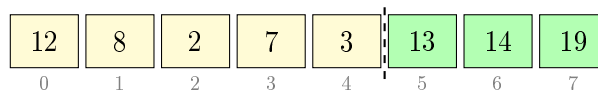
Schritt 38: $a[0]=2 < a[2]=12$ (groesseres Kind), tausche



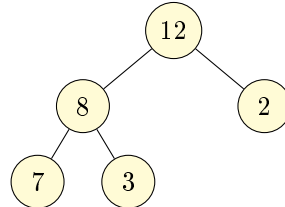
Schritt 39: $a[2]=2$ ist Blatt (keine Kinder), fertig



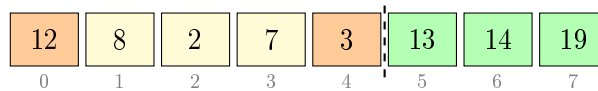
Schritt 40: Heap5: [12, 8, 2, 7, 3] | sortiert: [13, 14, 19]



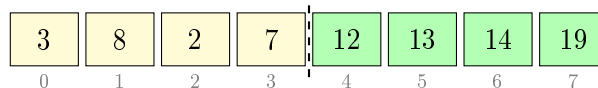
Baumdarstellung:



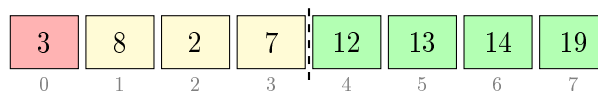
Schritt 41: Tausche Maximum $a[0]=12$ mit $a[4]=3$



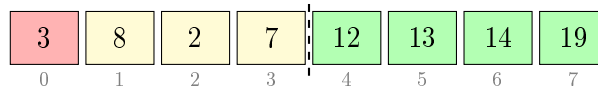
Schritt 42: $a[4]=12$ ist an Endposition, Heapgroesse: 4



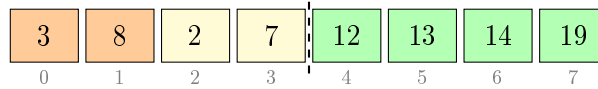
Schritt 43: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=3$



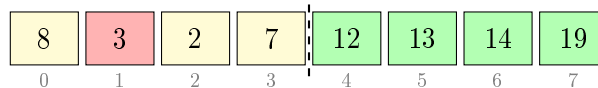
Schritt 44: Kinder: $a[1]=8$ und $a[2]=2$. Linkes Kind $8 \geq$ rechtes Kind 2, waehle linkes Kind



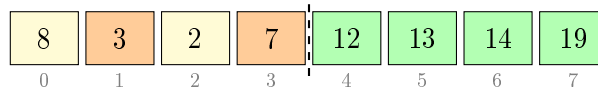
Schritt 45: $a[0]=3 < a[1]=8$ (großeres Kind), tausche



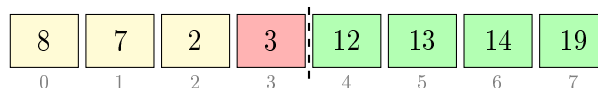
Schritt 46: Nur linkes Kind vorhanden: $a[3]=7$



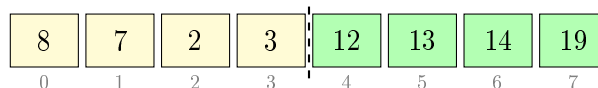
Schritt 47: $a[1]=3 < a[3]=7$ (großeres Kind), tausche



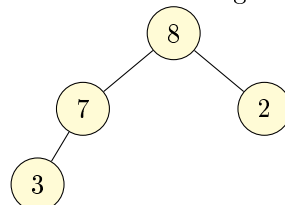
Schritt 48: $a[3]=3$ ist Blatt (keine Kinder), fertig



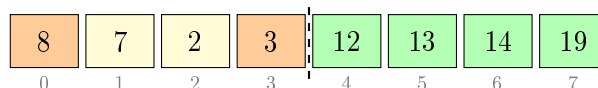
Schritt 49: Heap4: $[8, 7, 2, 3]$ | sortiert: $[12, 13, 14, 19]$



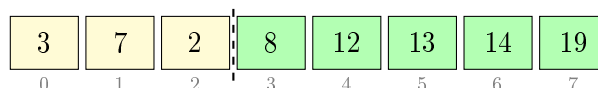
Baumdarstellung:



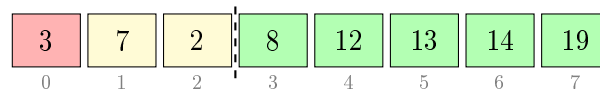
Schritt 50: Tausche Maximum $a[0]=8$ mit $a[3]=3$



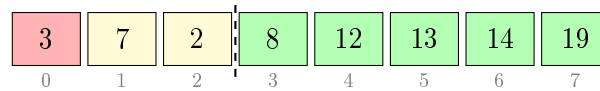
Schritt 51: $a[3]=8$ ist an Endposition, Heapgroesse: 3



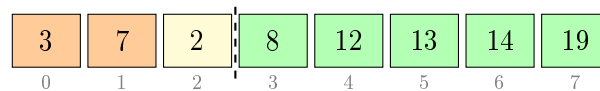
Schritt 52: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=3$



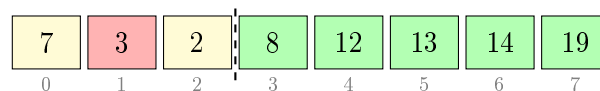
Schritt 53: Kinder: $a[1]=7$ und $a[2]=2$. Linkes Kind $7 \geq$ rechtes Kind 2, waehle linkes Kind



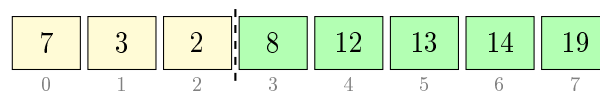
Schritt 54: $a[0]=3 < a[1]=7$ (groesseres Kind), tausche



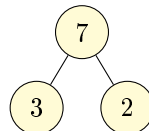
Schritt 55: $a[1]=3$ ist Blatt (keine Kinder), fertig



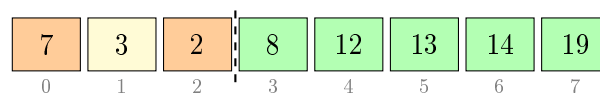
Schritt 56: Heap3: $[7, 3, 2]$ | sortiert: $[8, 12, 13, 14, 19]$



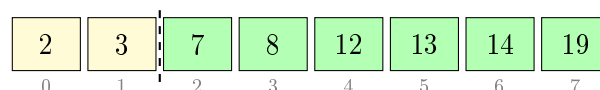
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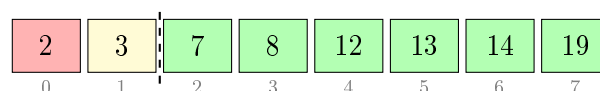
Schritt 57: Tausche Maximum $a[0]=7$ mit $a[2]=2$



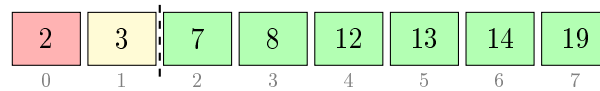
Schritt 58: $a[2]=7$ ist an Endposition, Heapgroesse: 2



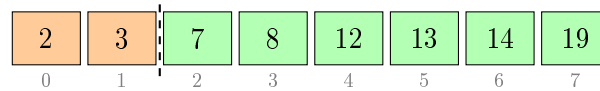
Schritt 59: $\text{siftdown}(a, 0, 2)$: Versickere $a[0]=2$



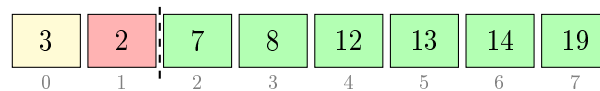
Schritt 60: Nur linkes Kind vorhanden: $a[1]=3$



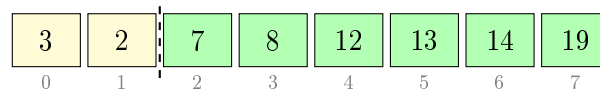
Schritt 61: $a[0]=2 < a[1]=3$ (grösseres Kind), tausche



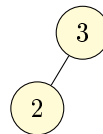
Schritt 62: $a[1]=2$ ist Blatt (keine Kinder), fertig



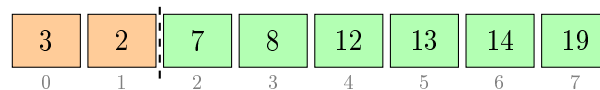
Schritt 63: Heap2: $[3, 2]$ | sortiert: $[7, 8, 12, 13, 14, 19]$



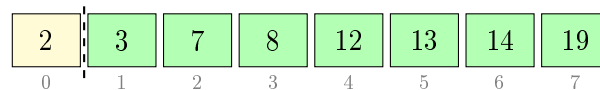
Baumdarstellung:



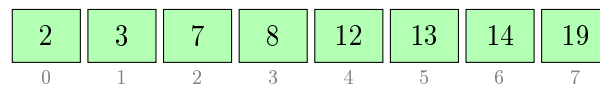
Schritt 64: Tausche Maximum $a[0]=3$ mit $a[1]=2$



Schritt 65: $a[1]=3$ ist an Endposition, Heapgrösse: 1



Schritt 66: Sortierung abgeschlossen: $[2, 3, 7, 8, 12, 13, 14, 19]$



Zusammenfassung der Heap-Zustände:

- **Heap8:** $[19, 13, 14, 7, 8, 12, 3, 2]$
- **Heap7:** $[14, 13, 12, 7, 8, 2, 3]$ | sortiert: $[19]$

- **Heap6:** [13, 8, 12, 7, 3, 2] | sortiert: [14, 19]
- **Heap5:** [12, 8, 2, 7, 3] | sortiert: [13, 14, 19]
- **Heap4:** [8, 7, 2, 3] | sortiert: [12, 13, 14, 19]
- **Heap3:** [7, 3, 2] | sortiert: [8, 12, 13, 14, 19]
- **Heap2:** [3, 2] | sortiert: [7, 8, 12, 13, 14, 19]

Antworten:

1. Anfangsheap (Heap8): **[19, 13, 14, 7, 8, 12, 3, 2]**

2. Zwischenheaps:

- Heap7: [14, 13, 12, 7, 8, 2, 3]
- Heap6: [13, 8, 12, 7, 3, 2]
- Heap5: [12, 8, 2, 7, 3]
- Heap4: [8, 7, 2, 3]
- Heap3: [7, 3, 2]
- Heap2: [3, 2]

3. Anzahl Vertauschungen: **16**

Endergebnis: [2, 3, 7, 8, 12, 13, 14, 19]

Aufgabe 2 – Schwierigkeit: ★★

Aufbau eines Anfangsheaps

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

2, 1, 5, 3, 4, 8, 7, 6

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

Fragen:

1. Geben Sie den Anfangsheap (Heap8) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap7, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgeführt?

Loesung:

Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

=== Phase 1: Anfangsheap herstellen ===

Schritt 1: $\text{siftdown}(a, 4, 8)$: Versickere $a[4]=4$

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

Schritt 2: $a[4]=4$ ist Blatt (keine Kinder), fertig

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

Schritt 3: $\text{siftdown}(a, 3, 8)$: Versickere $a[3]=3$

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

Schritt 4: Nur linkes Kind vorhanden: $a[7]=6$

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

Schritt 5: $a[3]=3 < a[7]=6$ (groesseres Kind), tausche

2	1	5	3	4	8	7	6
0	1	2	3	4	5	6	7

Schritt 6: $a[7]=3$ ist Blatt (keine Kinder), fertig

2	1	5	6	4	8	7	3
0	1	2	3	4	5	6	7

Schritt 7: $\text{siftdown}(a, 2, 8)$: Versickere $a[2]=5$

2	1	5	6	4	8	7	3
0	1	2	3	4	5	6	7

Schritt 8: Kinder: $a[5]=8$ und $a[6]=7$. Linkes Kind $8 \geq$ rechtes Kind 7, waehle linkes Kind

2	1	5	6	4	8	7	3
0	1	2	3	4	5	6	7

Schritt 9: $a[2]=5 < a[5]=8$ (groesseres Kind), tausche

2	1	5	6	4	8	7	3
0	1	2	3	4	5	6	7

Schritt 10: $a[5]=5$ ist Blatt (keine Kinder), fertig

2	1	8	6	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 11: $\text{siftdown}(a, 1, 8)$: Versickere $a[1]=1$

2	1	8	6	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 12: Kinder: $a[3]=6$ und $a[4]=4$. Linkes Kind $6 \geq$ rechtes Kind 4, waehle linkes Kind

2	1	8	6	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 13: $a[1]=1 < a[3]=6$ (groesseres Kind), tausche

2	1	8	6	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 14: Nur linkes Kind vorhanden: $a[7]=3$

2	6	8	1	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 15: $a[3]=1 < a[7]=3$ (groesseres Kind), tausche

2	6	8	1	4	5	7	3
0	1	2	3	4	5	6	7

Schritt 16: $a[7]=1$ ist Blatt (keine Kinder), fertig

2	6	8	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 17: $\text{siftdown}(a, 0, 8)$: Versickere $a[0]=2$

2	6	8	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 18: Kinder: $a[1]=6$ und $a[2]=8$. Rechtes Kind $8 >$ linkes Kind 6, waehle rechtes Kind

2	6	8	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 19: $a[0]=2 < a[2]=8$ (groesseres Kind), tausche

2	6	8	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 20: Kinder: $a[5]=5$ und $a[6]=7$. Rechtes Kind $7 >$ linkes Kind 5, waehle rechtes Kind

8	6	2	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 21: $a[2]=2 < a[6]=7$ (groesseres Kind), tausche

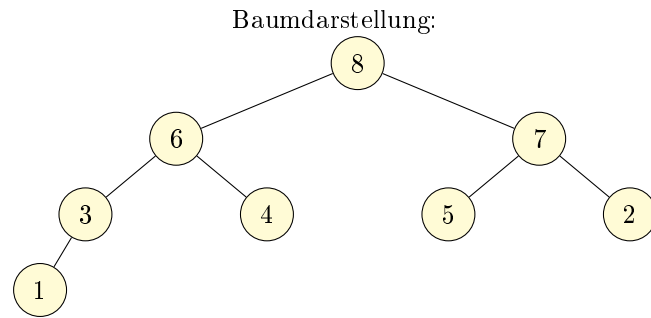
8	6	2	3	4	5	7	1
0	1	2	3	4	5	6	7

Schritt 22: $a[6]=2$ ist Blatt (keine Kinder), fertig

8	6	7	3	4	5	2	1
0	1	2	3	4	5	6	7

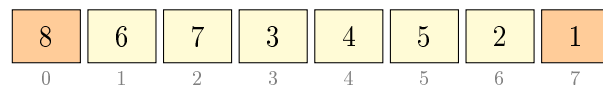
Schritt 23: Heap8: Anfangsheap hergestellt

8	6	7	3	4	5	2	1
0	1	2	3	4	5	6	7

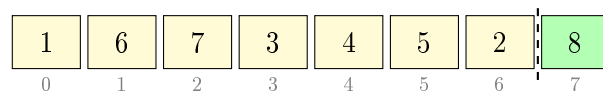


=== Phase 2: Sortieren durch Auswahl ===

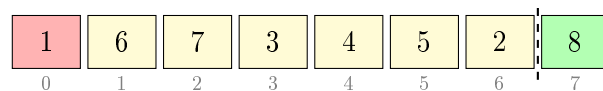
Schritt 24: Tausche Maximum $a[0]=8$ mit $a[7]=1$



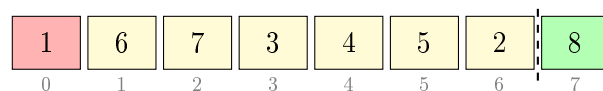
Schritt 25: $a[7]=8$ ist an Endposition, Heapgroesse: 7



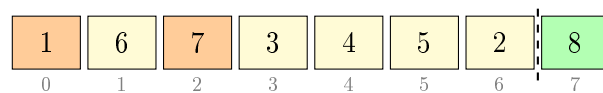
Schritt 26: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=1$



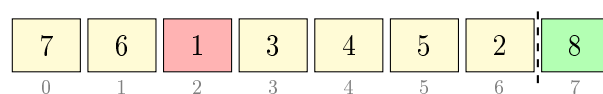
Schritt 27: Kinder: $a[1]=6$ und $a[2]=7$. Rechtes Kind $7 >$ linkes Kind 6, waehle rechtes Kind



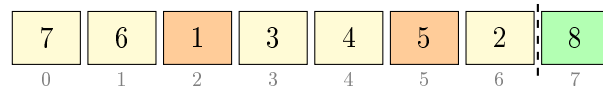
Schritt 28: $a[0]=1 < a[2]=7$ (groesseres Kind), tausche



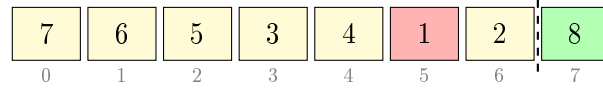
Schritt 29: Kinder: $a[5]=5$ und $a[6]=2$. Linkes Kind $5 \geq$ rechtes Kind 2, waehle linkes Kind



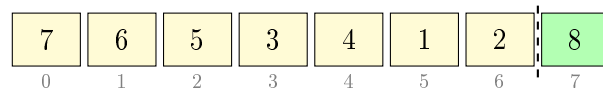
Schritt 30: $a[2]=1 < a[5]=5$ (groesseres Kind), tausche



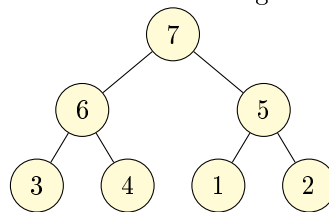
Schritt 31: $a[5]=1$ ist Blatt (keine Kinder), fertig



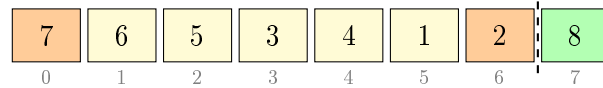
Schritt 32: Heap7: [7, 6, 5, 3, 4, 1, 2] | sortiert: [8]



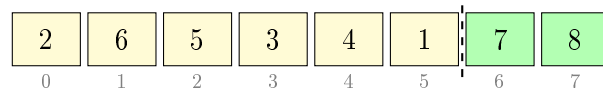
Baumdarstellung:



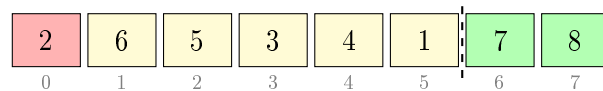
Schritt 33: Tausche Maximum $a[0]=7$ mit $a[6]=2$



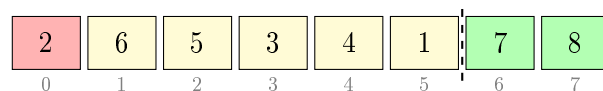
Schritt 34: $a[6]=7$ ist an Endposition, Heapgroesse: 6



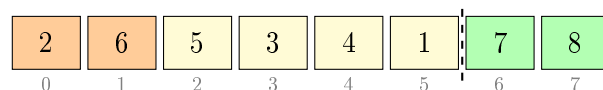
Schritt 35: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=2$



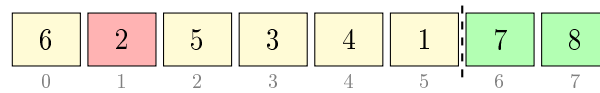
Schritt 36: Kinder: $a[1]=6$ und $a[2]=5$. Linkes Kind $6 \geq$ rechtes Kind 5 , wachle linkes Kind



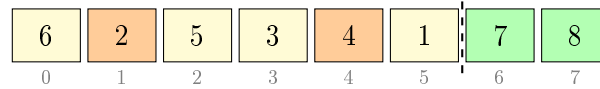
Schritt 37: $a[0]=2 < a[1]=6$ (groesseres Kind), tausche



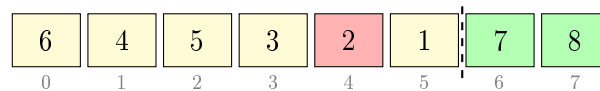
Schritt 38: Kinder: $a[3]=3$ und $a[4]=4$. Rechtes Kind $4 >$ linkes Kind 3 , waehle rechtes Kind



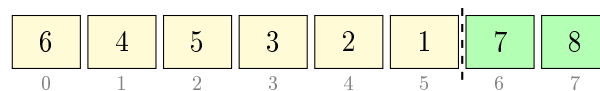
Schritt 39: $a[1]=2 < a[4]=4$ (groesseres Kind), tausche



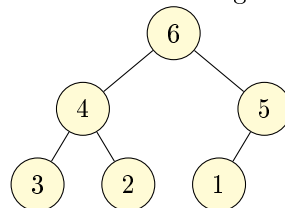
Schritt 40: $a[4]=2$ ist Blatt (keine Kinder), fertig



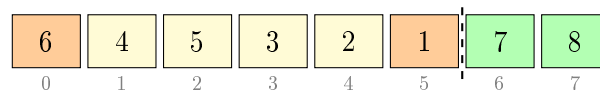
Schritt 41: Heap6: $[6, 4, 5, 3, 2, 1]$ | sortiert: $[7, 8]$



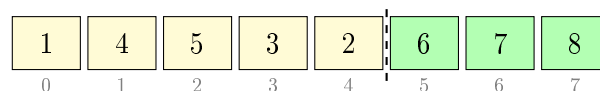
Baumdarstellung:



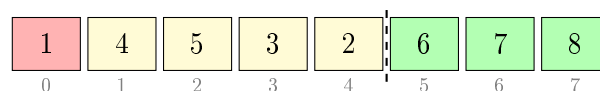
Schritt 42: Tausche Maximum $a[0]=6$ mit $a[5]=1$



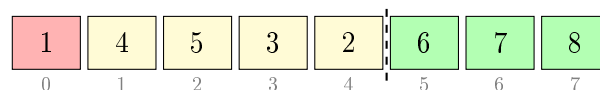
Schritt 43: $a[5]=6$ ist an Endposition, Heapgroesse: 5



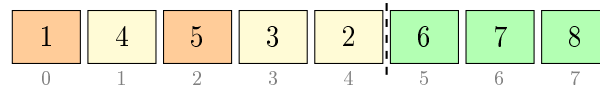
Schritt 44: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=1$



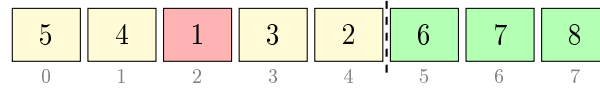
Schritt 45: Kinder: $a[1]=4$ und $a[2]=5$. Rechtes Kind $5 >$ linkes Kind 4 , waehle rechtes Kind



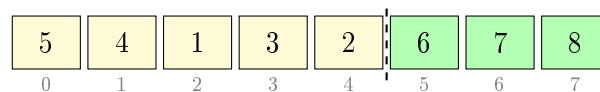
Schritt 46: $a[0]=1 < a[2]=5$ (grösseres Kind), tausche



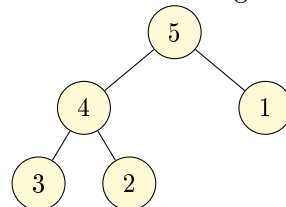
Schritt 47: $a[2]=1$ ist Blatt (keine Kinder), fertig



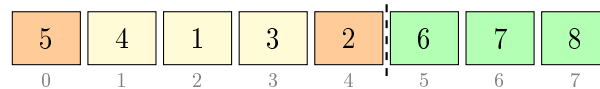
Schritt 48: Heap5: [5, 4, 1, 3, 2] | sortiert: [6, 7, 8]



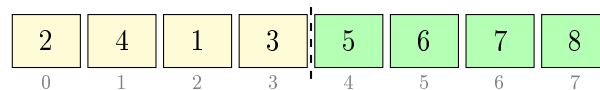
Baumdarstellung:



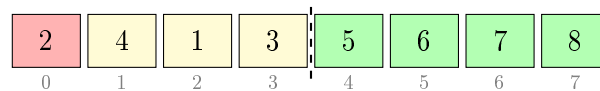
Schritt 49: Tausche Maximum $a[0]=5$ mit $a[4]=2$



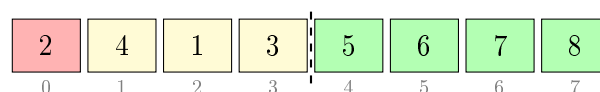
Schritt 50: $a[4]=5$ ist an Endposition, Heapgrösse: 4



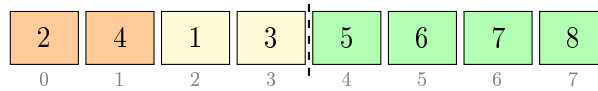
Schritt 51: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=2$



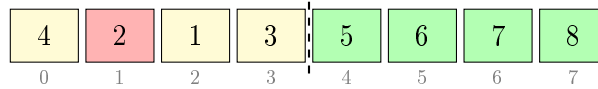
Schritt 52: Kinder: $a[1]=4$ und $a[2]=1$. Linkes Kind $4 \geq$ rechtes Kind 1, wähle linkes Kind



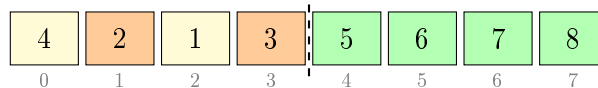
Schritt 53: $a[0]=2 < a[1]=4$ (grösseres Kind), tausche



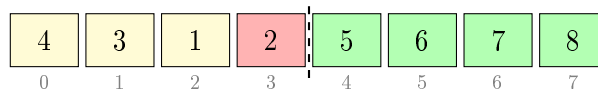
Schritt 54: Nur linkes Kind vorhanden: $a[3]=3$



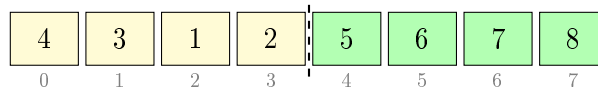
Schritt 55: $a[1]=2 < a[3]=3$ (grösseres Kind), tausche



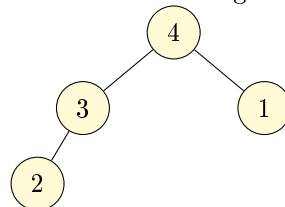
Schritt 56: $a[3]=2$ ist Blatt (keine Kinder), fertig



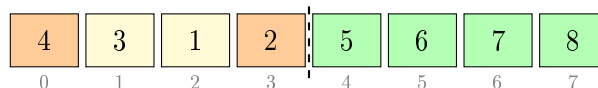
Schritt 57: Heap4: [4, 3, 1, 2] | sortiert: [5, 6, 7, 8]



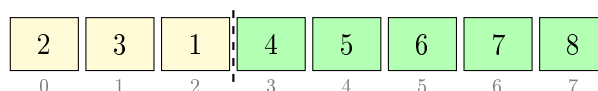
Baumdarstellung:



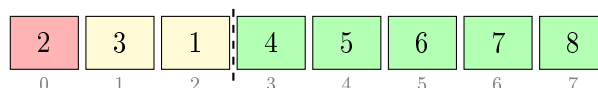
Schritt 58: Tausche Maximum $a[0]=4$ mit $a[3]=2$



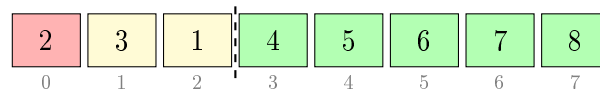
Schritt 59: $a[3]=4$ ist an Endposition, Heapgrösse: 3



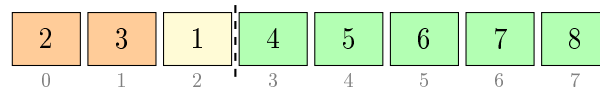
Schritt 60: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=2$



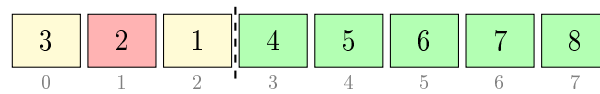
Schritt 61: Kinder: $a[1]=3$ und $a[2]=1$. Linkes Kind $3 \geq$ rechtes Kind 1, wähle linkes Kind



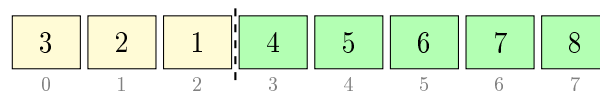
Schritt 62: $a[0]=2 < a[1]=3$ (grösseres Kind), tausche



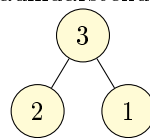
Schritt 63: $a[1]=2$ ist Blatt (keine Kinder), fertig



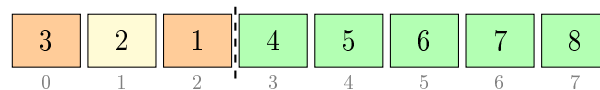
Schritt 64: Heap3: [3, 2, 1] | sortiert: [4, 5, 6, 7, 8]



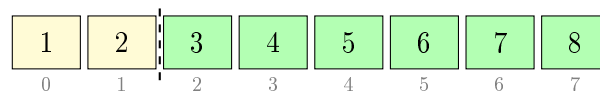
Baumdarstellung:



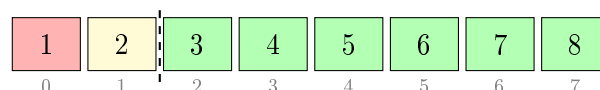
Schritt 65: Tausche Maximum $a[0]=3$ mit $a[2]=1$



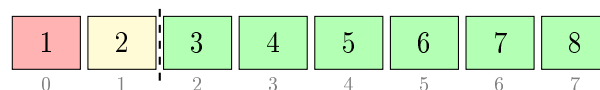
Schritt 66: $a[2]=3$ ist an Endposition, Heapgrösse: 2



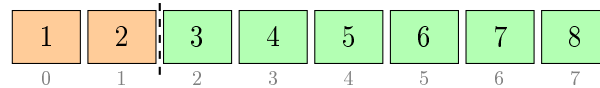
Schritt 67: $\text{siftdown}(a, 0, 2)$: Versickere $a[0]=1$



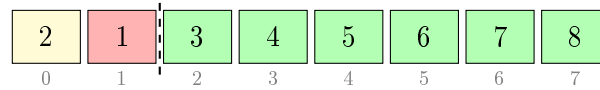
Schritt 68: Nur linkes Kind vorhanden: $a[1]=2$



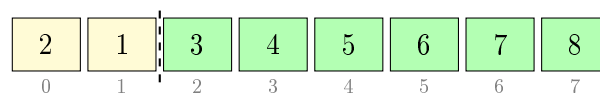
Schritt 69: $a[0]=1 < a[1]=2$ (großeres Kind), tausche



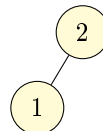
Schritt 70: $a[1]=1$ ist Blatt (keine Kinder), fertig



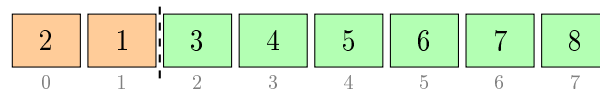
Schritt 71: Heap2: [2, 1] | sortiert: [3, 4, 5, 6, 7, 8]



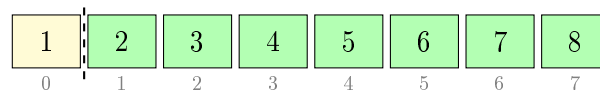
Baumdarstellung:



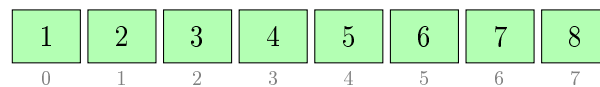
Schritt 72: Tausche Maximum $a[0]=2$ mit $a[1]=1$



Schritt 73: $a[1]=2$ ist an Endposition, Heapgröße: 1



Schritt 74: Sortierung abgeschlossen: [1, 2, 3, 4, 5, 6, 7, 8]



Zusammenfassung der Heap-Zustände:

- **Heap8:** [8, 6, 7, 3, 4, 5, 2, 1]
- **Heap7:** [7, 6, 5, 3, 4, 1, 2] | sortiert: [8]
- **Heap6:** [6, 4, 5, 3, 2, 1] | sortiert: [7, 8]
- **Heap5:** [5, 4, 1, 3, 2] | sortiert: [6, 7, 8]
- **Heap4:** [4, 3, 1, 2] | sortiert: [5, 6, 7, 8]

- **Heap3:** [3, 2, 1] | sortiert: [4, 5, 6, 7, 8]
- **Heap2:** [2, 1] | sortiert: [3, 4, 5, 6, 7, 8]

Antworten:

1. Anfangsheap (Heap8): [8, 6, 7, 3, 4, 5, 2, 1]
2. Zwischenheaps:
 - Heap7: [7, 6, 5, 3, 4, 1, 2]
 - Heap6: [6, 4, 5, 3, 2, 1]
 - Heap5: [5, 4, 1, 3, 2]
 - Heap4: [4, 3, 1, 2]
 - Heap3: [3, 2, 1]
 - Heap2: [2, 1]

3. Anzahl Vertauschungen: **22**

Endergebnis: [1, 2, 3, 4, 5, 6, 7, 8]

Aufgabe 3 – Schwierigkeit: ★★

Standard-Heapsort

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

5, 12, 3, 8, 1, 9, 4, 7

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

Fragen:

1. Geben Sie den Anfangsheap (Heap8) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap7, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgefuehrt?

Loesung:

Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

=== Phase 1: Anfangsheap herstellen ===

Schritt 1: $\text{siftdown}(a, 4, 8)$: Versickere $a[4]=1$

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 2: $a[4]=1$ ist Blatt (keine Kinder), fertig

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 3: $\text{siftdown}(a, 3, 8)$: Versickere $a[3]=8$

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 4: Nur linkes Kind vorhanden: $a[7]=7$

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 5: $a[3]=8 > a[7]=7$ (groesseres Kind), Heap-Bedingung erfuehlt

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 6: $\text{siftdown}(a, 2, 8)$: Versickere $a[2]=3$

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 7: Kinder: $a[5]=9$ und $a[6]=4$. Linkes Kind $9 \geq$ rechtes Kind 4, wähle linkes Kind

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 8: $a[2]=3 < a[5]=9$ (grösseres Kind), tausche

5	12	3	8	1	9	4	7
0	1	2	3	4	5	6	7

Schritt 9: $a[5]=3$ ist Blatt (keine Kinder), fertig

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

Schritt 10: $\text{siftdown}(a, 1, 8)$: Versickere $a[1]=12$

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

Schritt 11: Kinder: $a[3]=8$ und $a[4]=1$. Linkes Kind $8 \geq$ rechtes Kind 1, wähle linkes Kind

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

Schritt 12: $a[1]=12 \geq a[3]=8$ (grösseres Kind), Heap-Bedingung erfüllt

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

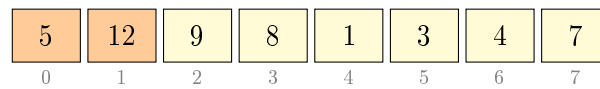
Schritt 13: $\text{siftdown}(a, 0, 8)$: Versickere $a[0]=5$

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

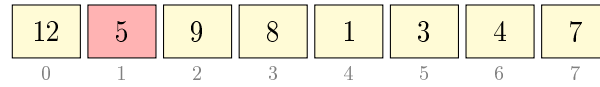
Schritt 14: Kinder: $a[1]=12$ und $a[2]=9$. Linkes Kind $12 \geq$ rechtes Kind 9, wähle linkes Kind

5	12	9	8	1	3	4	7
0	1	2	3	4	5	6	7

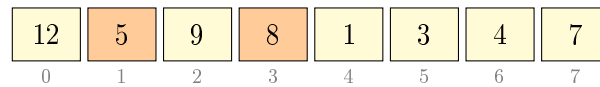
Schritt 15: $a[0]=5 < a[1]=12$ (groesseres Kind), tausche



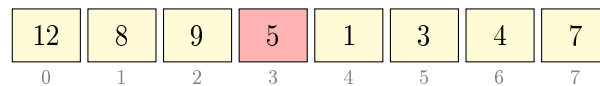
Schritt 16: Kinder: $a[3]=8$ und $a[4]=1$. Linkes Kind $8 \geq$ rechtes Kind 1, waehle linkes Kind



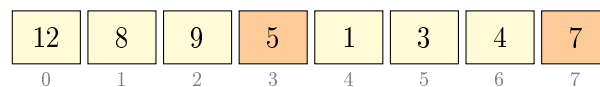
Schritt 17: $a[1]=5 < a[3]=8$ (groesseres Kind), tausche



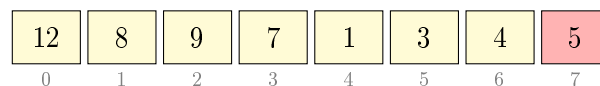
Schritt 18: Nur linkes Kind vorhanden: $a[7]=7$



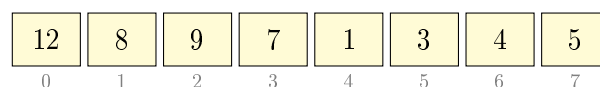
Schritt 19: $a[3]=5 < a[7]=7$ (groesseres Kind), tausche



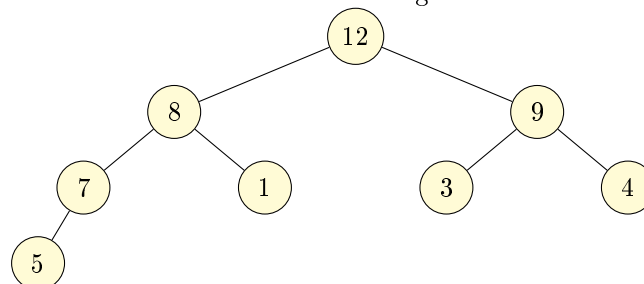
Schritt 20: $a[7]=5$ ist Blatt (keine Kinder), fertig



Schritt 21: Heap8: Anfangsheap hergestellt

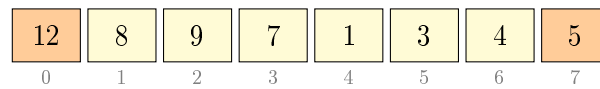


Baumdarstellung:

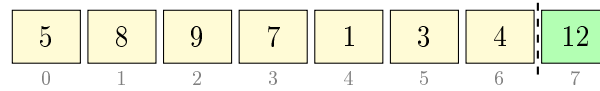


=== Phase 2: Sortieren durch Auswahl ===

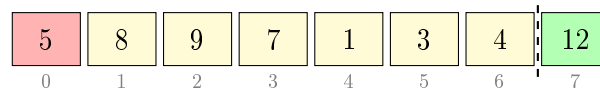
Schritt 22: Tausche Maximum $a[0]=12$ mit $a[7]=5$



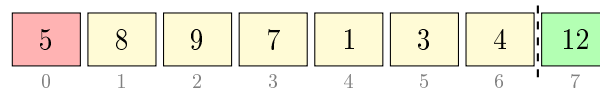
Schritt 23: $a[7]=12$ ist an Endposition, Heapgroesse: 7



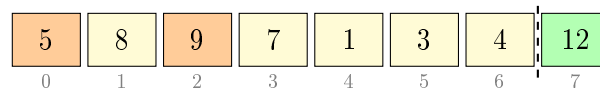
Schritt 24: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=5$



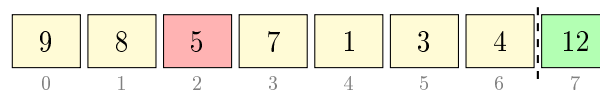
Schritt 25: Kinder: $a[1]=8$ und $a[2]=9$. Rechtes Kind $9 >$ linkes Kind 8, wachle rechtes Kind



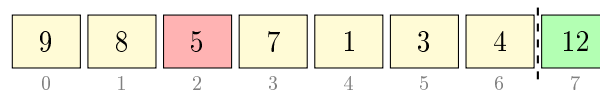
Schritt 26: $a[0]=5 < a[2]=9$ (groesseres Kind), tausche



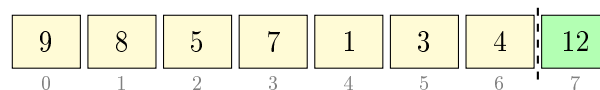
Schritt 27: Kinder: $a[5]=3$ und $a[6]=4$. Rechtes Kind $4 >$ linkes Kind 3, wachle rechtes Kind



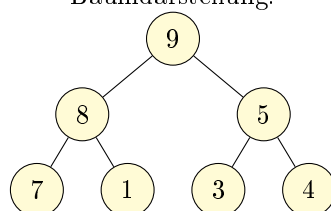
Schritt 28: $a[2]=5 \geq a[6]=4$ (groesseres Kind), Heap-Bedingung erfuehlt



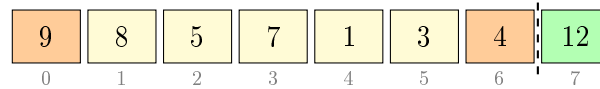
Schritt 29: Heap7: $[9, 8, 5, 7, 1, 3, 4]$ | sortiert: $[12]$



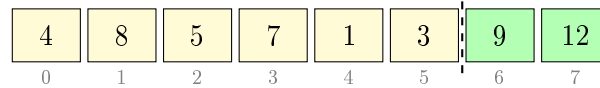
Baumdarstellung:



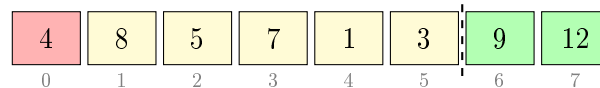
Schritt 30: Tausche Maximum $a[0]=9$ mit $a[6]=4$



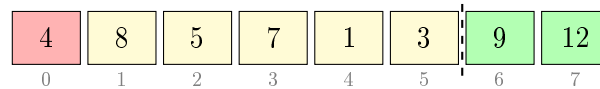
Schritt 31: $a[6]=9$ ist an Endposition, Heapgroesse: 6



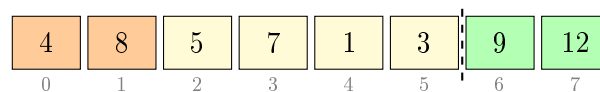
Schritt 32: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=4$



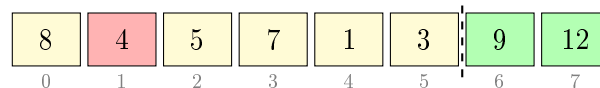
Schritt 33: Kinder: $a[1]=8$ und $a[2]=5$. Linkes Kind $8 \geq$ rechtes Kind 5, wähle linkes Kind



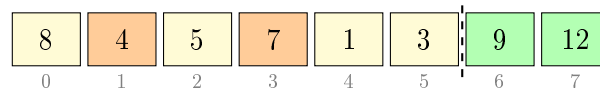
Schritt 34: $a[0]=4 < a[1]=8$ (grösseres Kind), tausche



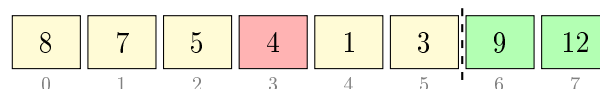
Schritt 35: Kinder: $a[3]=7$ und $a[4]=1$. Linkes Kind $7 \geq$ rechtes Kind 1, wähle linkes Kind



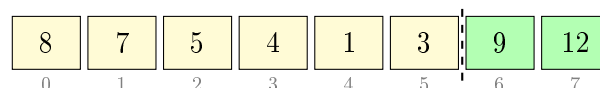
Schritt 36: $a[1]=4 < a[3]=7$ (grösseres Kind), tausche



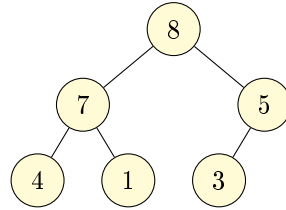
Schritt 37: $a[3]=4$ ist Blatt (keine Kinder), fertig



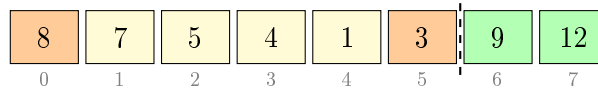
Schritt 38: Heap6: $[8, 7, 5, 4, 1, 3]$ | sortiert: $[9, 12]$



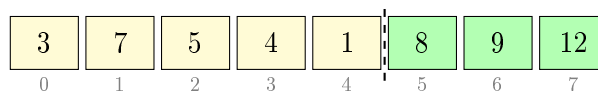
Baumdarstellung:



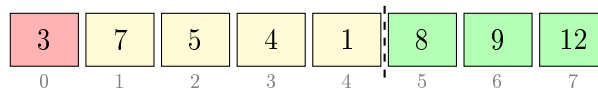
Schritt 39: Tausche Maximum $a[0]=8$ mit $a[5]=3$



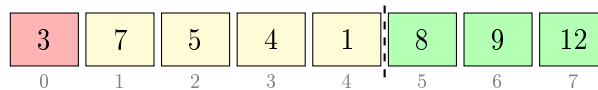
Schritt 40: $a[5]=8$ ist an Endposition, Heapgrösse: 5



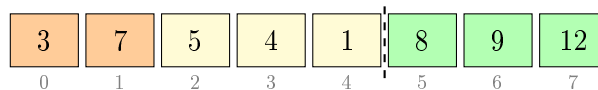
Schritt 41: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=3$



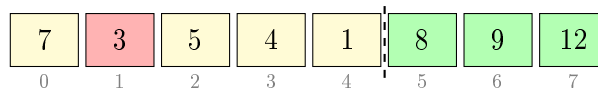
Schritt 42: Kinder: $a[1]=7$ und $a[2]=5$. Linkes Kind $7 \geq$ rechtes Kind 5, wähle linkes Kind



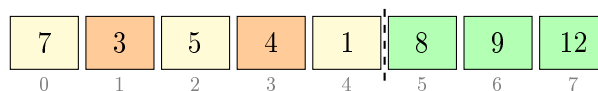
Schritt 43: $a[0]=3 < a[1]=7$ (grösseres Kind), tausche



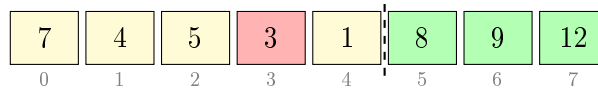
Schritt 44: Kinder: $a[3]=4$ und $a[4]=1$. Linkes Kind $4 \geq$ rechtes Kind 1, wähle linkes Kind



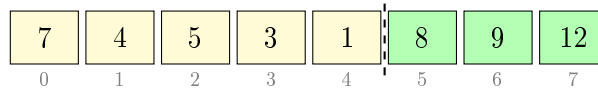
Schritt 45: $a[1]=3 < a[3]=4$ (grösseres Kind), tausche



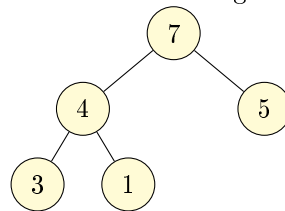
Schritt 46: $a[3]=3$ ist Blatt (keine Kinder), fertig



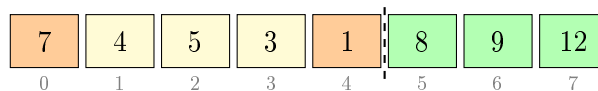
Schritt 47: Heap5: [7, 4, 5, 3, 1] | sortiert: [8, 9, 12]



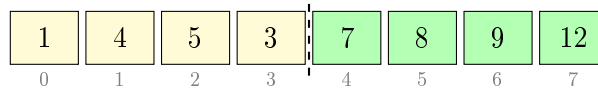
Baumdarstellung:



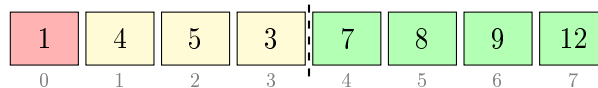
Schritt 48: Tausche Maximum $a[0]=7$ mit $a[4]=1$



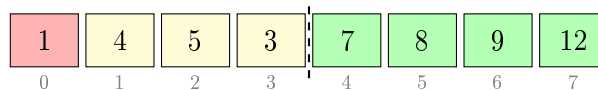
Schritt 49: $a[4]=7$ ist an Endposition, Heapgroesse: 4



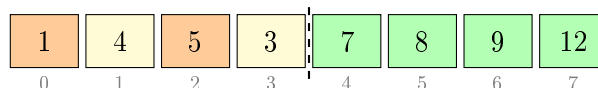
Schritt 50: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=1$



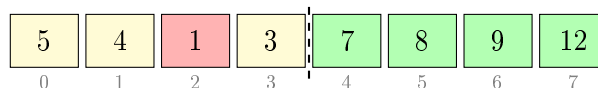
Schritt 51: Kinder: $a[1]=4$ und $a[2]=5$. Rechtes Kind $5 >$ linkes Kind 4, waehle rechtes Kind



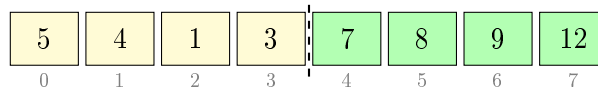
Schritt 52: $a[0]=1 < a[2]=5$ (groesseres Kind), tausche



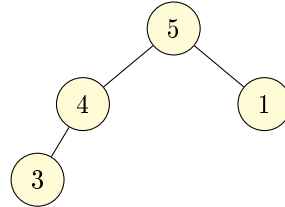
Schritt 53: $a[2]=1$ ist Blatt (keine Kinder), fertig



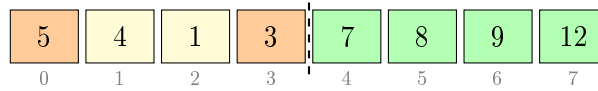
Schritt 54: Heap4: [5, 4, 1, 3] | sortiert: [7, 8, 9, 12]



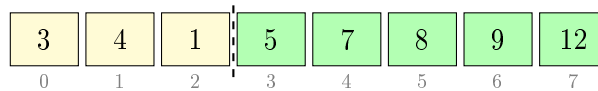
Baumdarstellung:



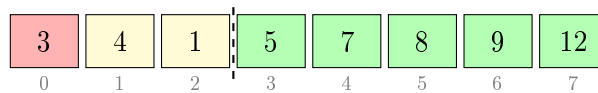
Schritt 55: Tausche Maximum $a[0]=5$ mit $a[3]=3$



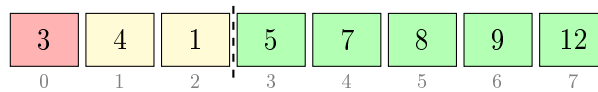
Schritt 56: $a[3]=5$ ist an Endposition, Heapgroesse: 3



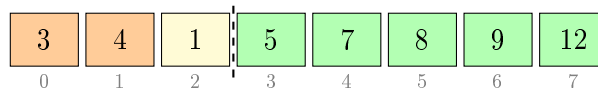
Schritt 57: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=3$



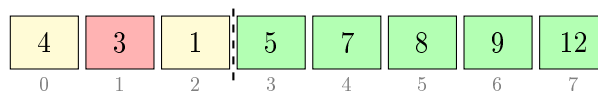
Schritt 58: Kinder: $a[1]=4$ und $a[2]=1$. Linkes Kind $4 \geq$ rechtes Kind 1, wachle linkes Kind



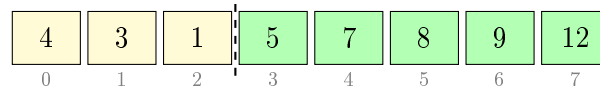
Schritt 59: $a[0]=3 < a[1]=4$ (groesseres Kind), tausche



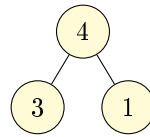
Schritt 60: $a[1]=3$ ist Blatt (keine Kinder), fertig



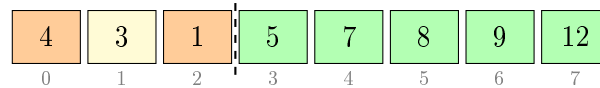
Schritt 61: Heap3: [4, 3, 1] | sortiert: [5, 7, 8, 9, 12]



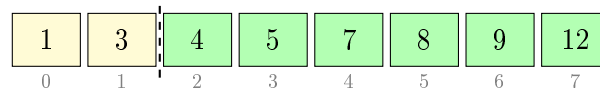
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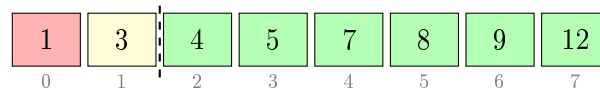
Schritt 62: Tausche Maximum $a[0]=4$ mit $a[2]=1$



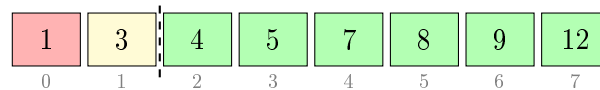
Schritt 63: $a[2]=4$ ist an Endposition, Heapgröße: 2



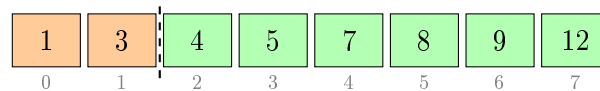
Schritt 64: $\text{siftdown}(a, 0, 2)$: Versickere $a[0]=1$



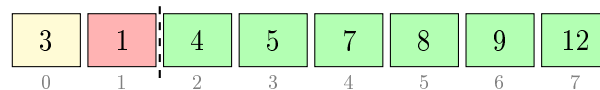
Schritt 65: Nur linkes Kind vorhanden: $a[1]=3$



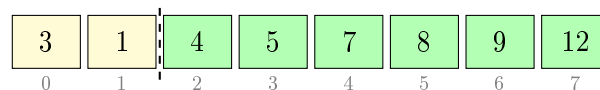
Schritt 66: $a[0]=1 < a[1]=3$ (großeres Kind), tausche



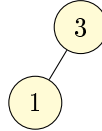
Schritt 67: $a[1]=1$ ist Blatt (keine Kinder), fertig



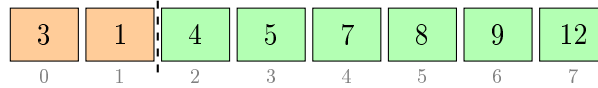
Schritt 68: Heap2: $[3, 1]$ | sortiert: $[4, 5, 7, 8, 9, 12]$



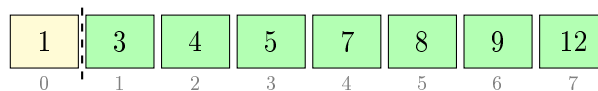
Baumdarstellung:



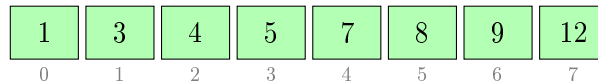
Schritt 69: Tausche Maximum $a[0]=3$ mit $a[1]=1$



Schritt 70: $a[1]=3$ ist an Endposition, Heapgroesse: 1



Schritt 71: Sortierung abgeschlossen: $[1, 3, 4, 5, 7, 8, 9, 12]$



Zusammenfassung der Heap-Zustände:

- **Heap8:** $[12, 8, 9, 7, 1, 3, 4, 5]$
- **Heap7:** $[9, 8, 5, 7, 1, 3, 4]$ | sortiert: $[12]$
- **Heap6:** $[8, 7, 5, 4, 1, 3]$ | sortiert: $[9, 12]$
- **Heap5:** $[7, 4, 5, 3, 1]$ | sortiert: $[8, 9, 12]$
- **Heap4:** $[5, 4, 1, 3]$ | sortiert: $[7, 8, 9, 12]$
- **Heap3:** $[4, 3, 1]$ | sortiert: $[5, 7, 8, 9, 12]$
- **Heap2:** $[3, 1]$ | sortiert: $[4, 5, 7, 8, 9, 12]$

Antworten:

1. Anfangsheap (Heap8): **$[12, 8, 9, 7, 1, 3, 4, 5]$**

2. Zwischenheaps:

- Heap7: $[9, 8, 5, 7, 1, 3, 4]$
- Heap6: $[8, 7, 5, 4, 1, 3]$
- Heap5: $[7, 4, 5, 3, 1]$
- Heap4: $[5, 4, 1, 3]$
- Heap3: $[4, 3, 1]$
- Heap2: $[3, 1]$

3. Anzahl Vertauschungen: **19**

Endergebnis: $[1, 3, 4, 5, 7, 8, 9, 12]$

Aufgabe 4 – Schwierigkeit: ★★★

Heapsort mit 10 Elementen

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

15, 8, 23, 4, 16, 42, 7, 29, 11, 3

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

Fragen:

1. Geben Sie den Anfangsheap (Heap10) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap9, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgeführt?

Loesung:

Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

=== Phase 1: Anfangsheap herstellen ===

Schritt 1: siftdown(a, 5, 10): Versickere a[5]=42

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 2: a[5]=42 ist Blatt (keine Kinder), fertig

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 3: siftdown(a, 4, 10): Versickere a[4]=16

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 4: Nur linkes Kind vorhanden: a[9]=3

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 5: a[4]=16 >= a[9]=3 (groesseres Kind), Heap-Bedingung erfuehlt

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 6: siftdown(a, 3, 10): Versickere $a[3]=4$

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 7: Kinder: $a[7]=29$ und $a[8]=11$. Linkes Kind $29 \geq$ rechtes Kind 11, waehle linkes Kind

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 8: $a[3]=4 < a[7]=29$ (groesseres Kind), tausche

15	8	23	4	16	42	7	29	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 9: $a[7]=4$ ist Blatt (keine Kinder), fertig

15	8	23	29	16	42	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 10: siftdown(a, 2, 10): Versickere $a[2]=23$

15	8	23	29	16	42	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 11: Kinder: $a[5]=42$ und $a[6]=7$. Linkes Kind $42 \geq$ rechtes Kind 7, waehle linkes Kind

15	8	23	29	16	42	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 12: $a[2]=23 < a[5]=42$ (groesseres Kind), tausche

15	8	23	29	16	42	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 13: $a[5]=23$ ist Blatt (keine Kinder), fertig

15	8	42	29	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 14: siftdown(a, 1, 10): Versickere $a[1]=8$

15	8	42	29	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 15: Kinder: $a[3]=29$ und $a[4]=16$. Linkes Kind $29 \geq$ rechtes Kind 16 , waehle linkes Kind

15	8	42	29	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 16: $a[1]=8 < a[3]=29$ (groesseres Kind), tausche

15	8	42	29	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 17: Kinder: $a[7]=4$ und $a[8]=11$. Rechtes Kind $11 >$ linkes Kind 4 , waehle rechtes Kind

15	29	42	8	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 18: $a[3]=8 < a[8]=11$ (groesseres Kind), tausche

15	29	42	8	16	23	7	4	11	3
0	1	2	3	4	5	6	7	8	9

Schritt 19: $a[8]=8$ ist Blatt (keine Kinder), fertig

15	29	42	11	16	23	7	4	8	3
0	1	2	3	4	5	6	7	8	9

Schritt 20: $\text{siftdown}(a, 0, 10)$: Versickere $a[0]=15$

15	29	42	11	16	23	7	4	8	3
0	1	2	3	4	5	6	7	8	9

Schritt 21: Kinder: $a[1]=29$ und $a[2]=42$. Rechtes Kind $42 >$ linkes Kind 29 , waehle rechtes Kind

15	29	42	11	16	23	7	4	8	3
0	1	2	3	4	5	6	7	8	9

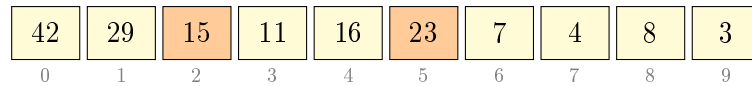
Schritt 22: $a[0]=15 < a[2]=42$ (groesseres Kind), tausche

15	29	42	11	16	23	7	4	8	3
0	1	2	3	4	5	6	7	8	9

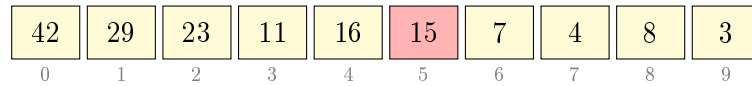
Schritt 23: Kinder: $a[5]=23$ und $a[6]=7$. Linkes Kind $23 \geq$ rechtes Kind 7 , waehle linkes Kind

42	29	15	11	16	23	7	4	8	3
0	1	2	3	4	5	6	7	8	9

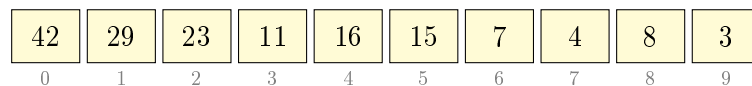
Schritt 24: $a[2]=15 < a[5]=23$ (groesseres Kind), tausche



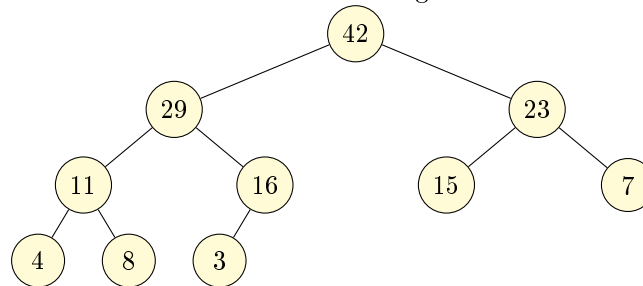
Schritt 25: $a[5]=15$ ist Blatt (keine Kinder), fertig



Schritt 26: Heap10: Anfangsheap hergestellt

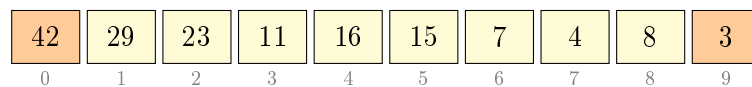


Baumdarstellung:

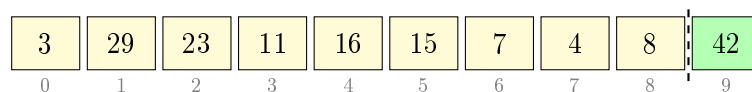


=== Phase 2: Sortieren durch Auswahl ===

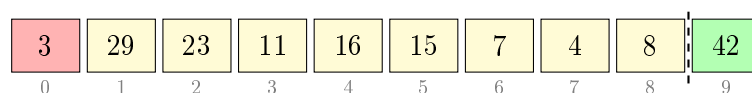
Schritt 27: Tausche Maximum $a[0]=42$ mit $a[9]=3$



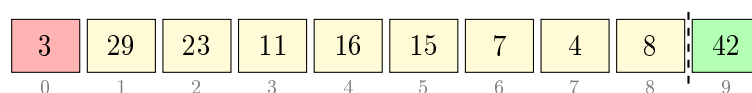
Schritt 28: $a[9]=42$ ist an Endposition, Heapgröße: 9



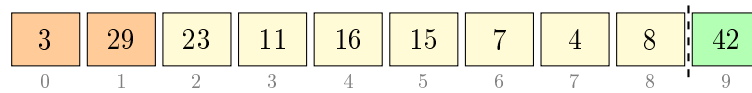
Schritt 29: $\text{siftdown}(a, 0, 9)$: Versickere $a[0]=3$



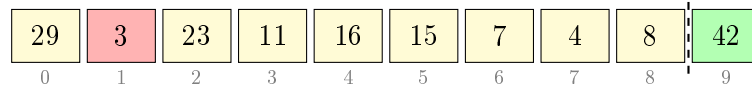
Schritt 30: Kinder: $a[1]=29$ und $a[2]=23$. Linkes Kind $29 \geq$ rechtes Kind 23 , wähle linkes Kind



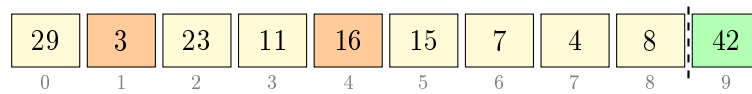
Schritt 31: $a[0]=3 < a[1]=29$ (groesseres Kind), tausche



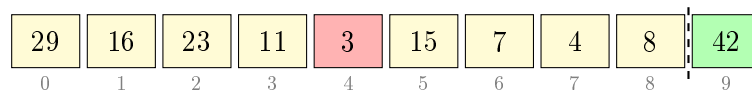
Schritt 32: Kinder: $a[3]=11$ und $a[4]=16$. Rechtes Kind $16 >$ linkes Kind 11 , waehle rechtes Kind



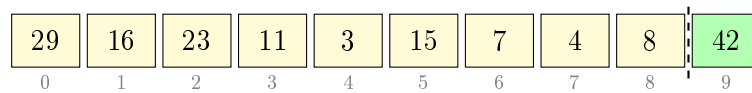
Schritt 33: $a[1]=3 < a[4]=16$ (groesseres Kind), tausche



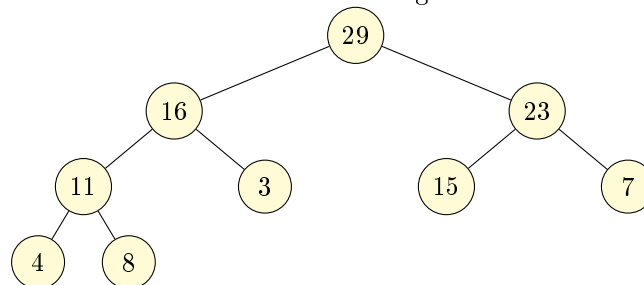
Schritt 34: $a[4]=3$ ist Blatt (keine Kinder), fertig



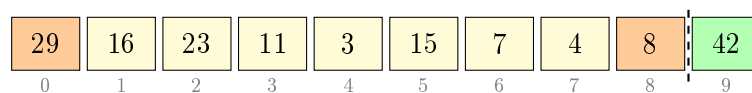
Schritt 35: Heap9: [29, 16, 23, 11, 3, 15, 7, 4, 8] | sortiert: [42]



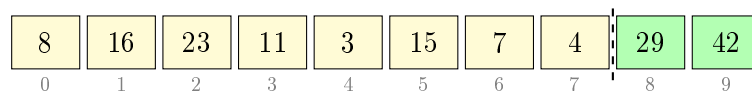
Baumdarstellung:



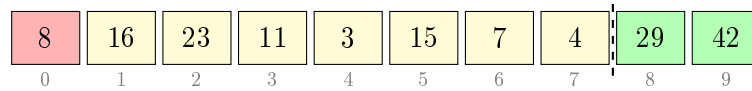
Schritt 36: Tausche Maximum $a[0]=29$ mit $a[8]=8$



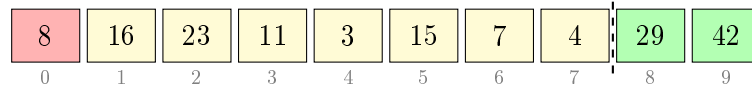
Schritt 37: $a[8]=29$ ist an Endposition, Heapgroesse: 8



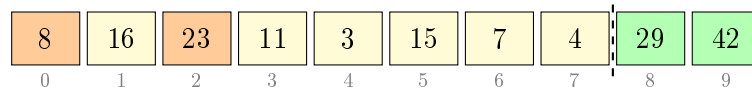
Schritt 38: $\text{siftdown}(a, 0, 8)$: Versickere $a[0]=8$



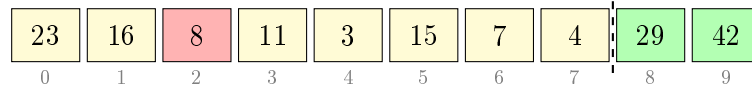
Schritt 39: Kinder: $a[1]=16$ und $a[2]=23$. Rechtes Kind $23 >$ linkes Kind 16 , waehle rechtes Kind



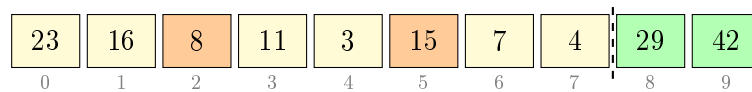
Schritt 40: $a[0]=8 < a[2]=23$ (groesseres Kind), tausche



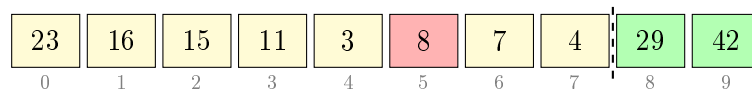
Schritt 41: Kinder: $a[5]=15$ und $a[6]=7$. Linkes Kind $15 \geq$ rechtes Kind 7 , waehle linkes Kind



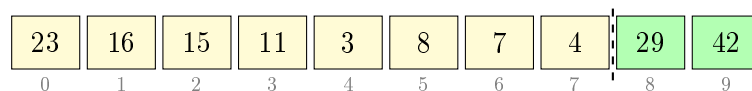
Schritt 42: $a[2]=8 < a[5]=15$ (groesseres Kind), tausche



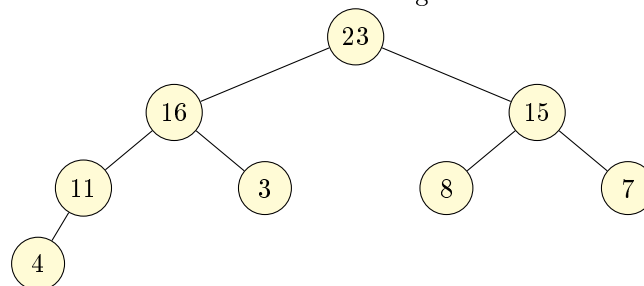
Schritt 43: $a[5]=8$ ist Blatt (keine Kinder), fertig



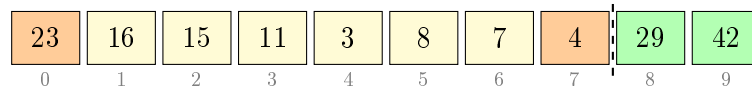
Schritt 44: Heap8: $[23, 16, 15, 11, 3, 8, 7, 4]$ | sortiert: $[29, 42]$



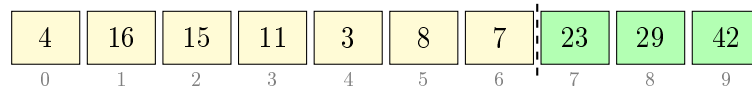
Baumdarstellung:



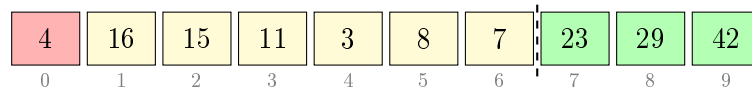
Schritt 45: Tausche Maximum $a[0]=23$ mit $a[7]=4$



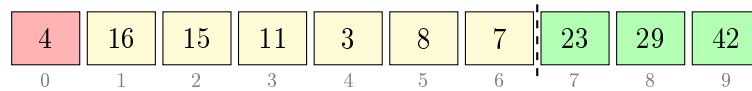
Schritt 46: $a[7]=23$ ist an Endposition, Heapgroesse: 7



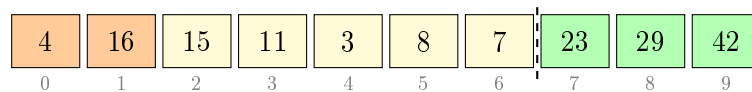
Schritt 47: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=4$



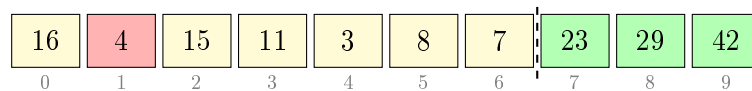
Schritt 48: Kinder: $a[1]=16$ und $a[2]=15$. Linkes Kind $16 \geq$ rechtes Kind 15, waehle linkes Kind



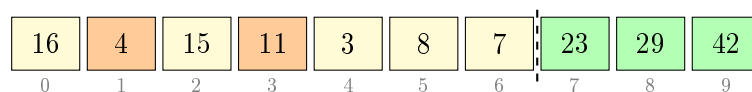
Schritt 49: $a[0]=4 < a[1]=16$ (groesseres Kind), tausche



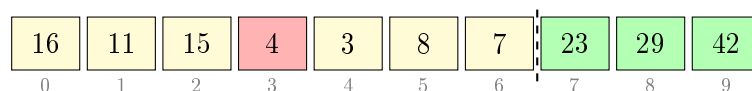
Schritt 50: Kinder: $a[3]=11$ und $a[4]=3$. Linkes Kind $11 \geq$ rechtes Kind 3, waehle linkes Kind



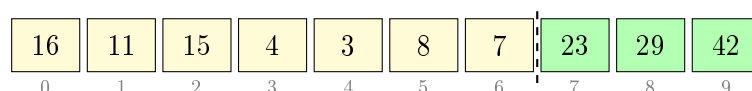
Schritt 51: $a[1]=4 < a[3]=11$ (groesseres Kind), tausche



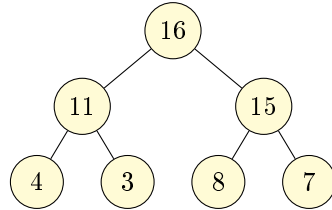
Schritt 52: $a[3]=4$ ist Blatt (keine Kinder), fertig



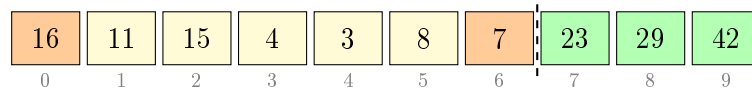
Schritt 53: Heap7: $[16, 11, 15, 4, 3, 8, 7]$ | sortiert: $[23, 29, 42]$



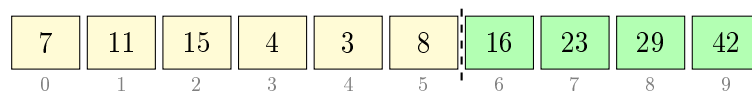
Baumdarstellung:



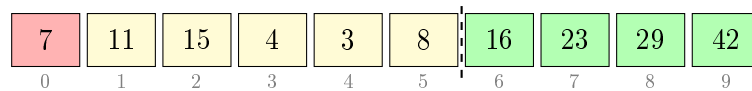
Schritt 54: Tausche Maximum $a[0]=16$ mit $a[6]=7$



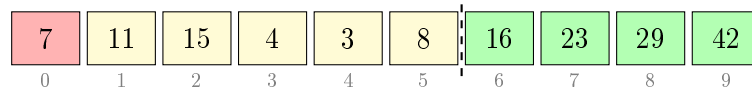
Schritt 55: $a[6]=16$ ist an Endposition, Heapgrösse: 6



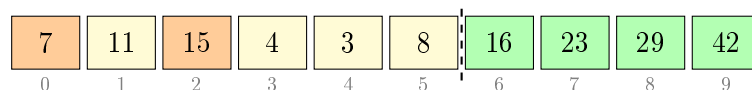
Schritt 56: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=7$



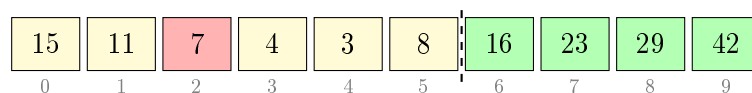
Schritt 57: Kinder: $a[1]=11$ und $a[2]=15$. Rechtes Kind $15 >$ linkes Kind 11 , wähle rechtes Kind



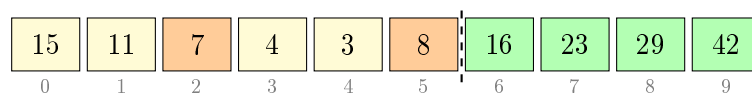
Schritt 58: $a[0]=7 < a[2]=15$ (grösseres Kind), tausche



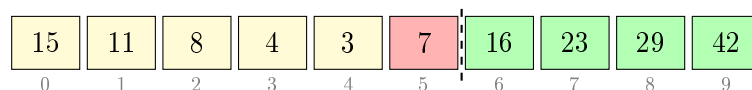
Schritt 59: Nur linkes Kind vorhanden: $a[5]=8$



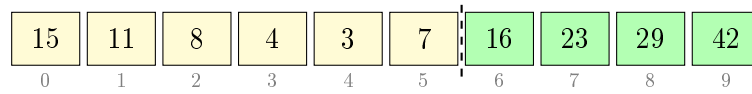
Schritt 60: $a[2]=7 < a[5]=8$ (grösseres Kind), tausche



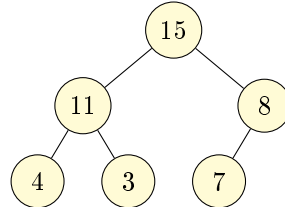
Schritt 61: $a[5]=7$ ist Blatt (keine Kinder), fertig



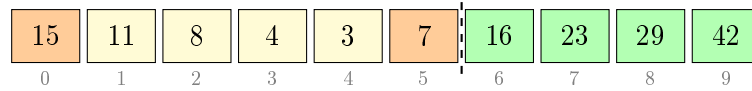
Schritt 62: Heap6: [15, 11, 8, 4, 3, 7] | sortiert: [16, 23, 29, 42]



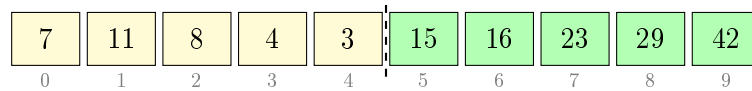
Baumdarstellung:



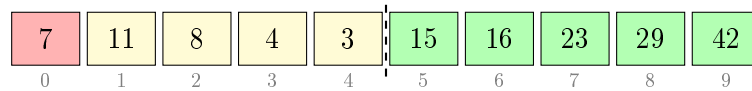
Schritt 63: Tausche Maximum $a[0]=15$ mit $a[5]=7$



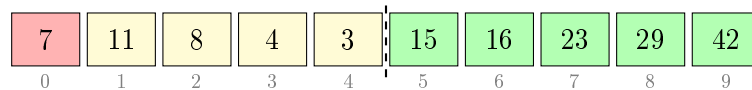
Schritt 64: $a[5]=15$ ist an Endposition, Heapgrösse: 5



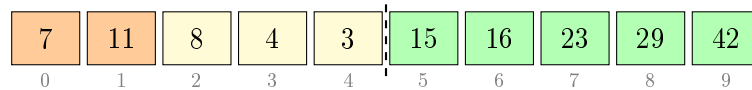
Schritt 65: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=7$



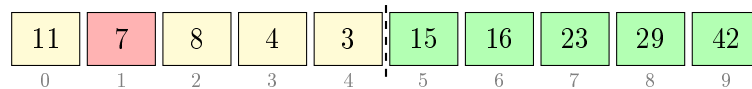
Schritt 66: Kinder: $a[1]=11$ und $a[2]=8$. Linkes Kind $11 \geq$ rechtes Kind 8, wähle linkes Kind



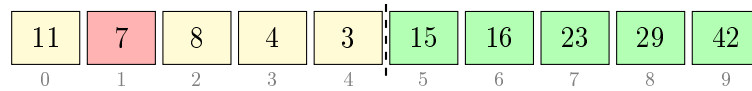
Schritt 67: $a[0]=7 < a[1]=11$ (grösseres Kind), tausche



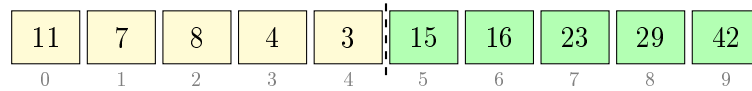
Schritt 68: Kinder: $a[3]=4$ und $a[4]=3$. Linkes Kind $4 \geq$ rechtes Kind 3, wähle linkes Kind



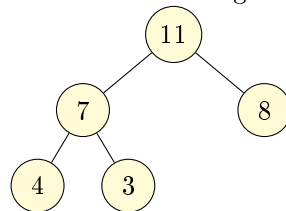
Schritt 69: $a[1]=7 \geq a[3]=4$ (grösseres Kind), Heap-Bedingung erfüllt



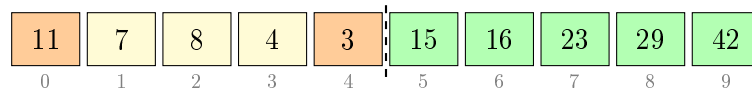
Schritt 70: Heap5: [11, 7, 8, 4, 3] | sortiert: [15, 16, 23, 29, 42]



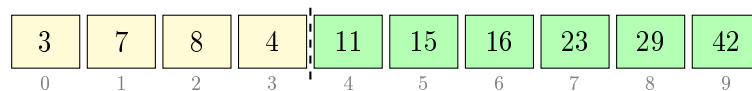
Baumdarstellung:



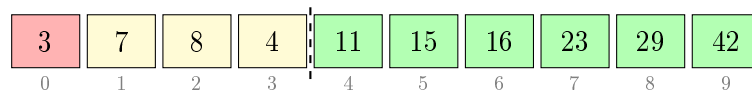
Schritt 71: Tausche Maximum $a[0]=11$ mit $a[4]=3$



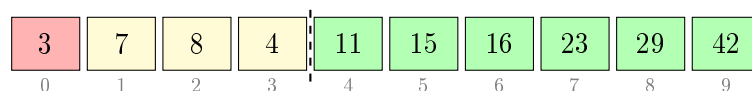
Schritt 72: $a[4]=11$ ist an Endposition, Heapgrösse: 4



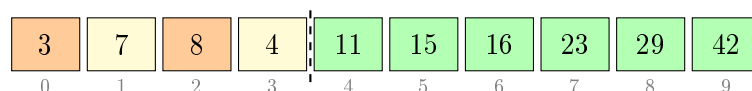
Schritt 73: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=3$



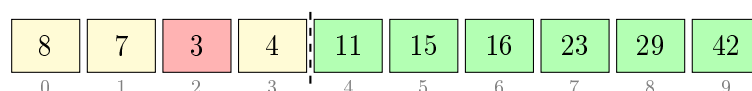
Schritt 74: Kinder: $a[1]=7$ und $a[2]=8$. Rechtes Kind $8 >$ linkes Kind 7, wähle rechtes Kind



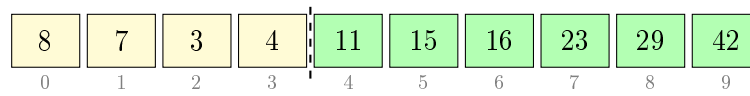
Schritt 75: $a[0]=3 < a[2]=8$ (grösseres Kind), tausche



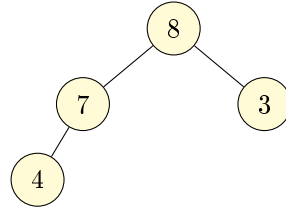
Schritt 76: $a[2]=3$ ist Blatt (keine Kinder), fertig



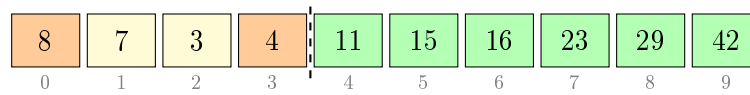
Schritt 77: Heap4: [8, 7, 3, 4] | sortiert: [11, 15, 16, 23, 29, 42]



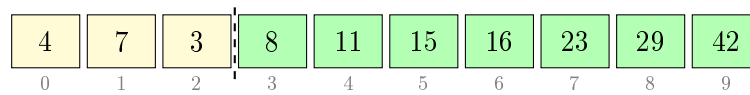
Baumdarstellung:



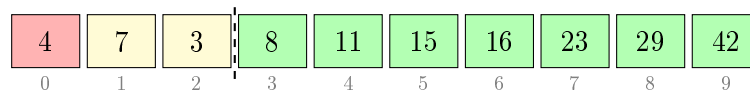
Schritt 78: Tausche Maximum $a[0]=8$ mit $a[3]=4$



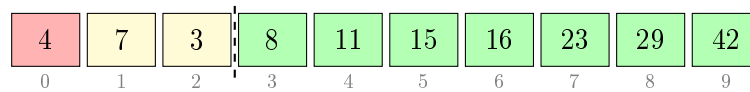
Schritt 79: $a[3]=8$ ist an Endposition, Heapgrösse: 3



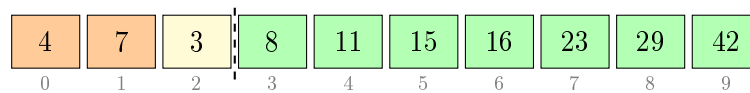
Schritt 80: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=4$



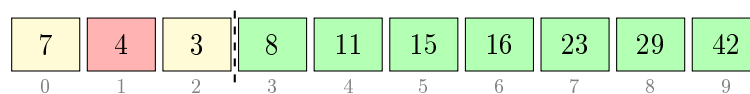
Schritt 81: Kinder: $a[1]=7$ und $a[2]=3$. Linkes Kind $7 \geq$ rechtes Kind 3, wähle linkes Kind



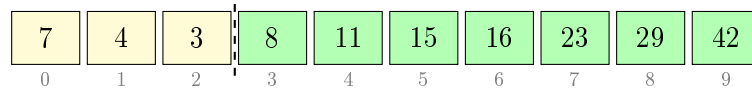
Schritt 82: $a[0]=4 < a[1]=7$ (grösseres Kind), tausche



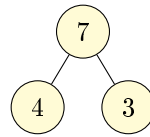
Schritt 83: $a[1]=4$ ist Blatt (keine Kinder), fertig



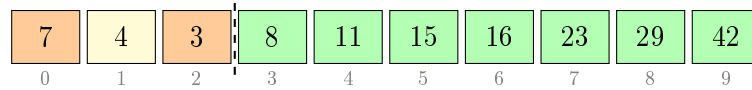
Schritt 84: Heap3: [7, 4, 3] | sortiert: [8, 11, 15, 16, 23, 29, 42]



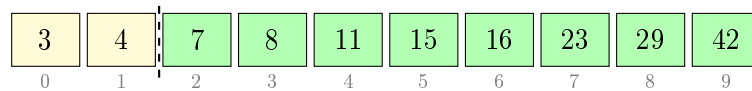
Baumdarstellung:



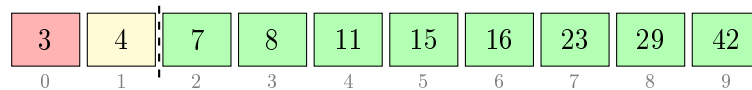
Schritt 85: Tausche Maximum $a[0]=7$ mit $a[2]=3$



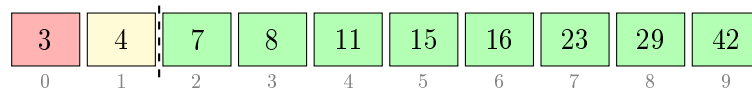
Schritt 86: $a[2]=7$ ist an Endposition, Heapgrösse: 2



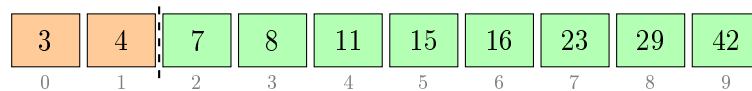
Schritt 87: $\text{siftdown}(a, 0, 2)$: Versickere $a[0]=3$



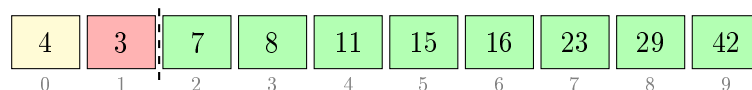
Schritt 88: Nur linkes Kind vorhanden: $a[1]=4$



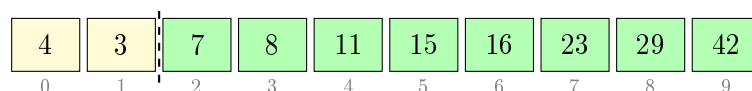
Schritt 89: $a[0]=3 < a[1]=4$ (grösseres Kind), tausche



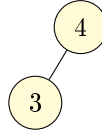
Schritt 90: $a[1]=3$ ist Blatt (keine Kinder), fertig



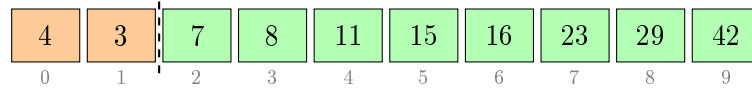
Schritt 91: Heap2: $[4, 3]$ | sortiert: $[7, 8, 11, 15, 16, 23, 29, 42]$



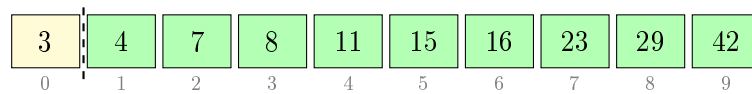
Baumdarstellung:



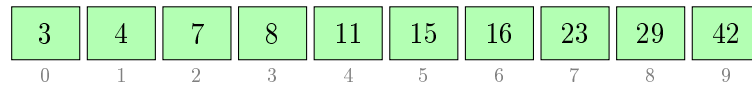
Schritt 92: Tausche Maximum $a[0]=4$ mit $a[1]=3$



Schritt 93: $a[1]=4$ ist an Endposition, Heapgröße: 1



Schritt 94: Sortierung abgeschlossen: [3, 4, 7, 8, 11, 15, 16, 23, 29, 42]



Zusammenfassung der Heap-Zustände:

- **Heap10:** [42, 29, 23, 11, 16, 15, 7, 4, 8, 3]
 - **Heap9:** [29, 16, 23, 11, 3, 15, 7, 4, 8] | sortiert: [42]
 - **Heap8:** [23, 16, 15, 11, 3, 8, 7, 4] | sortiert: [29, 42]
 - **Heap7:** [16, 11, 15, 4, 3, 8, 7] | sortiert: [23, 29, 42]
 - **Heap6:** [15, 11, 8, 4, 3, 7] | sortiert: [16, 23, 29, 42]
 - **Heap5:** [11, 7, 8, 4, 3] | sortiert: [15, 16, 23, 29, 42]
 - **Heap4:** [8, 7, 3, 4] | sortiert: [11, 15, 16, 23, 29, 42]
 - **Heap3:** [7, 4, 3] | sortiert: [8, 11, 15, 16, 23, 29, 42]
 - **Heap2:** [4, 3] | sortiert: [7, 8, 11, 15, 16, 23, 29, 42]
-

Antworten:

1. Anfangsheap (Heap10): [42, 29, 23, 11, 16, 15, 7, 4, 8, 3]
2. Zwischenheaps:
 - Heap9: [29, 16, 23, 11, 3, 15, 7, 4, 8]
 - Heap8: [23, 16, 15, 11, 3, 8, 7, 4]
 - Heap7: [16, 11, 15, 4, 3, 8, 7]
 - Heap6: [15, 11, 8, 4, 3, 7]

- Heap5: [11, 7, 8, 4, 3]
- Heap4: [8, 7, 3, 4]
- Heap3: [7, 4, 3]
- Heap2: [4, 3]

3. Anzahl Vertauschungen: **27**

Endergebnis: [3, 4, 7, 8, 11, 15, 16, 23, 29, 42]

Aufgabe 5 – Schwierigkeit: ★★★

Heapsort mit aufsteigend sortierter Eingabe

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

1, 2, 3, 4, 5, 6, 7, 8

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

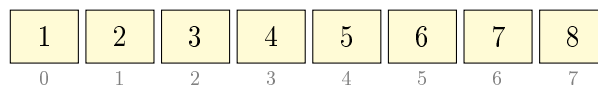
Fragen:

1. Geben Sie den Anfangsheap (Heap8) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap7, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgeführt?

Loesung:

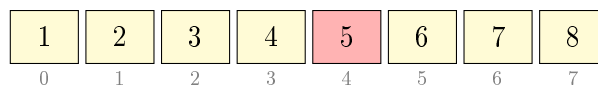
Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

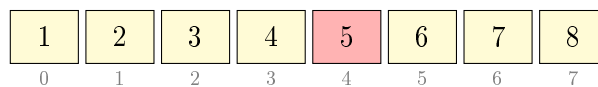


=== Phase 1: Anfangsheap herstellen ===

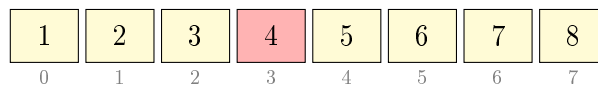
Schritt 1: $\text{siftdown}(a, 4, 8)$: Versickere $a[4]=5$



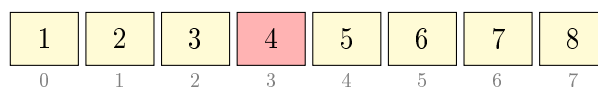
Schritt 2: $a[4]=5$ ist Blatt (keine Kinder), fertig



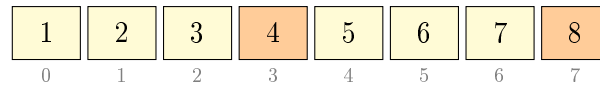
Schritt 3: $\text{siftdown}(a, 3, 8)$: Versickere $a[3]=4$



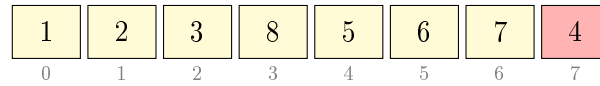
Schritt 4: Nur linkes Kind vorhanden: $a[7]=8$



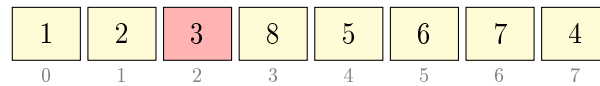
Schritt 5: $a[3]=4 < a[7]=8$ (groesseres Kind), tausche



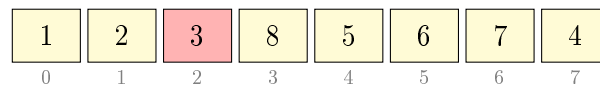
Schritt 6: $a[7]=4$ ist Blatt (keine Kinder), fertig



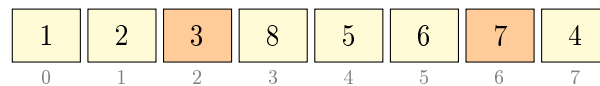
Schritt 7: $\text{siftdown}(a, 2, 8)$: Versickere $a[2]=3$



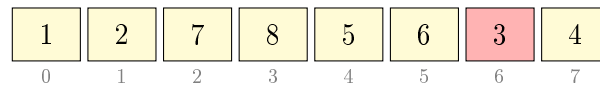
Schritt 8: Kinder: $a[5]=6$ und $a[6]=7$. Rechtes Kind $7 >$ linkes Kind 6, waehle rechtes Kind



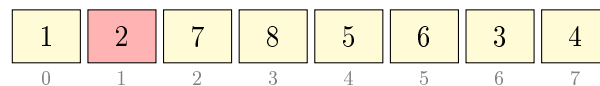
Schritt 9: $a[2]=3 < a[6]=7$ (groesseres Kind), tausche



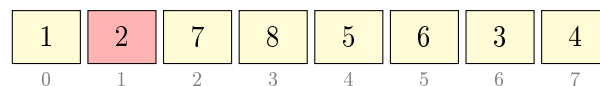
Schritt 10: $a[6]=3$ ist Blatt (keine Kinder), fertig



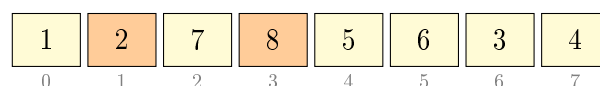
Schritt 11: $\text{siftdown}(a, 1, 8)$: Versickere $a[1]=2$



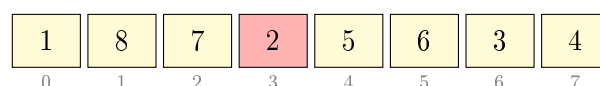
Schritt 12: Kinder: $a[3]=8$ und $a[4]=5$. Linkes Kind $8 \geq$ rechtes Kind 5, waehle linkes Kind



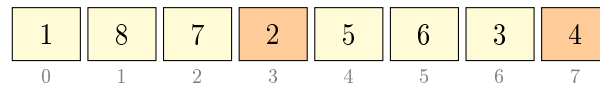
Schritt 13: $a[1]=2 < a[3]=8$ (groesseres Kind), tausche



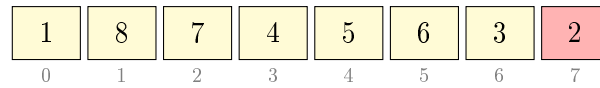
Schritt 14: Nur linkes Kind vorhanden: $a[7]=4$



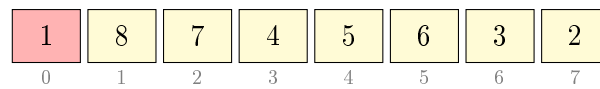
Schritt 15: $a[3]=2 < a[7]=4$ (groesseres Kind), tausche



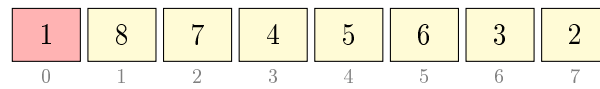
Schritt 16: $a[7]=2$ ist Blatt (keine Kinder), fertig



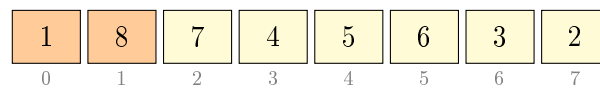
Schritt 17: $\text{siftdown}(a, 0, 8)$: Versickere $a[0]=1$



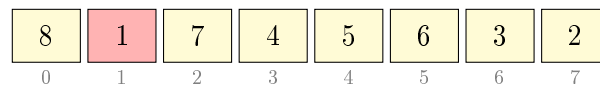
Schritt 18: Kinder: $a[1]=8$ und $a[2]=7$. Linkes Kind $8 \geq$ rechtes Kind 7, waehle linkes Kind



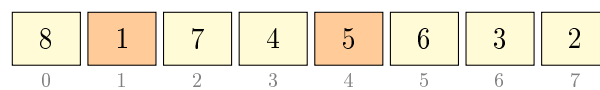
Schritt 19: $a[0]=1 < a[1]=8$ (groesseres Kind), tausche



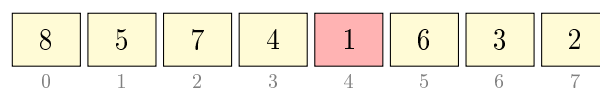
Schritt 20: Kinder: $a[3]=4$ und $a[4]=5$. Rechtes Kind $5 >$ linkes Kind 4, waehle rechtes Kind



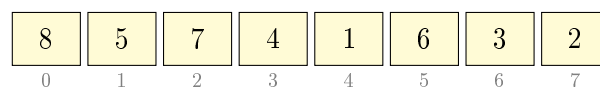
Schritt 21: $a[1]=1 < a[4]=5$ (groesseres Kind), tausche

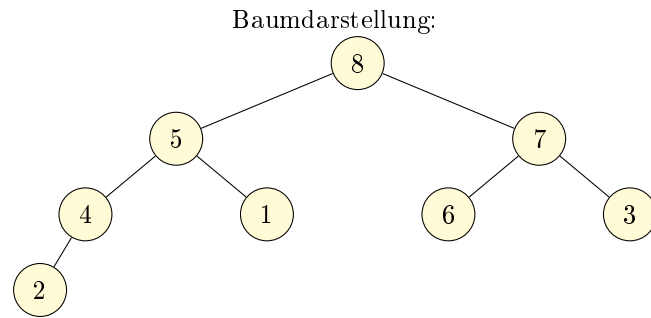


Schritt 22: $a[4]=1$ ist Blatt (keine Kinder), fertig



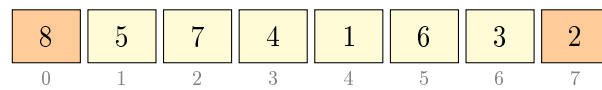
Schritt 23: Heap8: Anfangsheap hergestellt



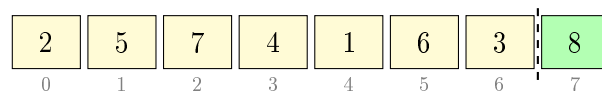


=== Phase 2: Sortieren durch Auswahl ===

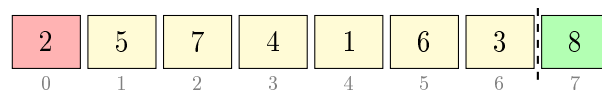
Schritt 24: Tausche Maximum $a[0]=8$ mit $a[7]=2$



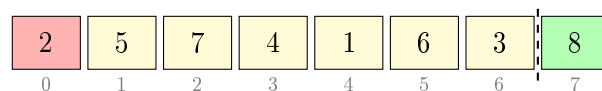
Schritt 25: $a[7]=8$ ist an Endposition, Heapgroesse: 7



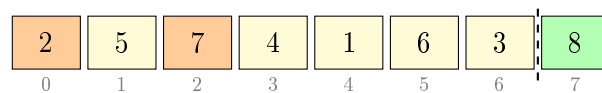
Schritt 26: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=2$



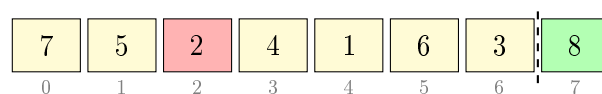
Schritt 27: Kinder: $a[1]=5$ und $a[2]=7$. Rechtes Kind $7 >$ linkes Kind 5, waehle rechtes Kind



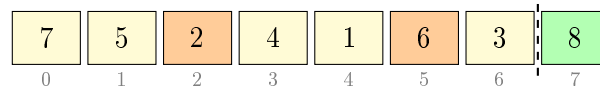
Schritt 28: $a[0]=2 < a[2]=7$ (groesseres Kind), tausche



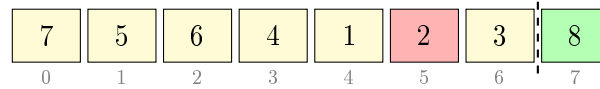
Schritt 29: Kinder: $a[5]=6$ und $a[6]=3$. Linkes Kind $6 \geq$ rechtes Kind 3, waehle linkes Kind



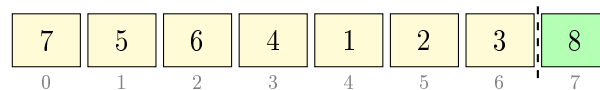
Schritt 30: $a[2]=2 < a[5]=6$ (groesseres Kind), tausche



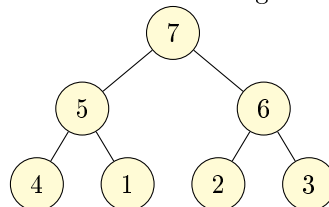
Schritt 31: $a[5]=2$ ist Blatt (keine Kinder), fertig



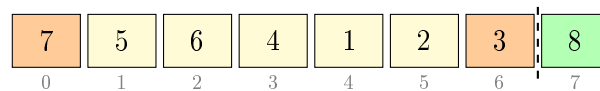
Schritt 32: Heap7: [7, 5, 6, 4, 1, 2, 3] | sortiert: [8]



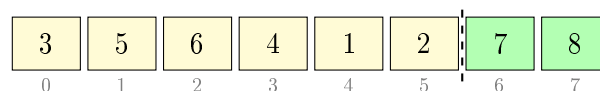
Baumdarstellung:



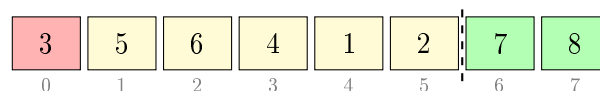
Schritt 33: Tausche Maximum $a[0]=7$ mit $a[6]=3$



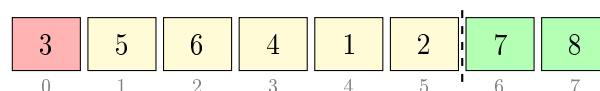
Schritt 34: $a[6]=7$ ist an Endposition, Heapgroesse: 6



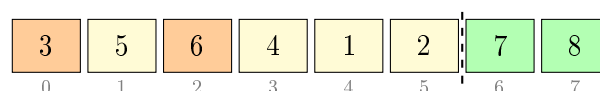
Schritt 35: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=3$



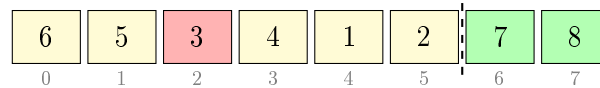
Schritt 36: Kinder: $a[1]=5$ und $a[2]=6$. Rechtes Kind $6 >$ linkes Kind 5, wähle rechtes Kind



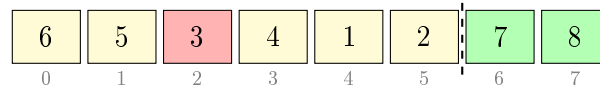
Schritt 37: $a[0]=3 < a[2]=6$ (grösseres Kind), tausche



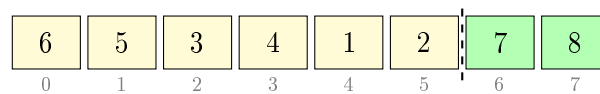
Schritt 38: Nur linkes Kind vorhanden: $a[5]=2$



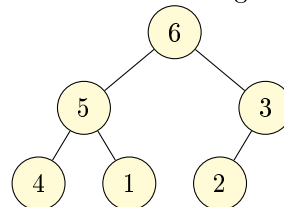
Schritt 39: $a[2]=3 \geq a[5]=2$ (grösseres Kind), Heap-Bedingung erfüllt



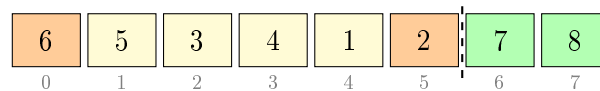
Schritt 40: Heap6: [6, 5, 3, 4, 1, 2] | sortiert: [7, 8]



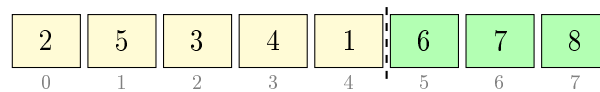
Baumdarstellung:



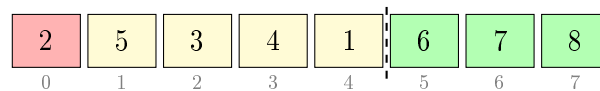
Schritt 41: Tausche Maximum $a[0]=6$ mit $a[5]=2$



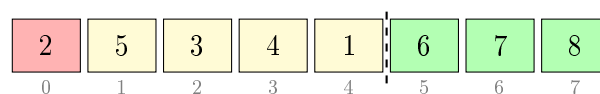
Schritt 42: $a[5]=6$ ist an Endposition, Heapgrösse: 5



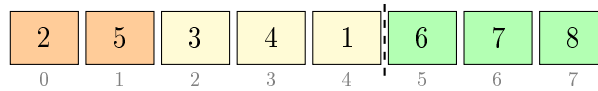
Schritt 43: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=2$



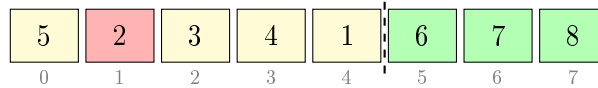
Schritt 44: Kinder: $a[1]=5$ und $a[2]=3$. Linkes Kind $5 \geq$ rechtes Kind 3 , wähle linkes Kind



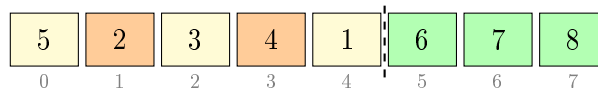
Schritt 45: $a[0]=2 < a[1]=5$ (grösseres Kind), tausche



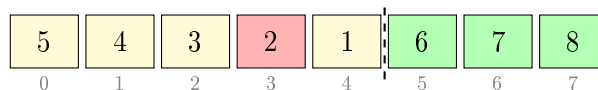
Schritt 46: Kinder: $a[3]=4$ und $a[4]=1$. Linkes Kind $4 \geq$ rechtes Kind 1, wähle linkes Kind



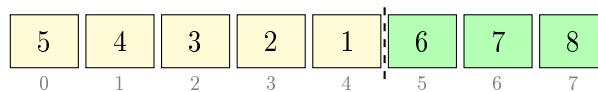
Schritt 47: $a[1]=2 < a[3]=4$ (grösseres Kind), tausche



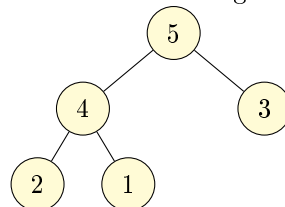
Schritt 48: $a[3]=2$ ist Blatt (keine Kinder), fertig



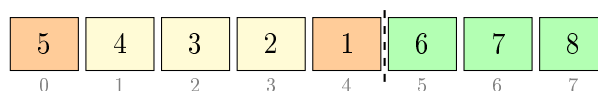
Schritt 49: Heap5: $[5, 4, 3, 2, 1]$ | sortiert: $[6, 7, 8]$



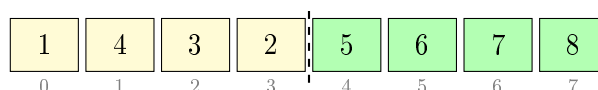
Baumdarstellung:



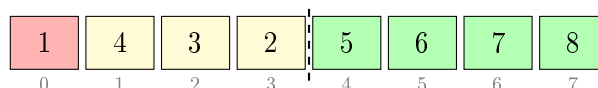
Schritt 50: Tausche Maximum $a[0]=5$ mit $a[4]=1$



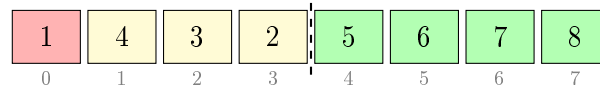
Schritt 51: $a[4]=5$ ist an Endposition, Heapgrösse: 4



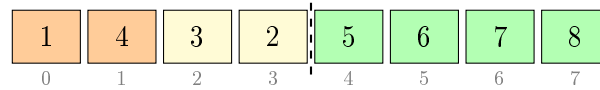
Schritt 52: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=1$



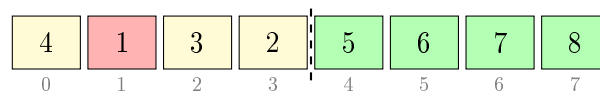
Schritt 53: Kinder: $a[1]=4$ und $a[2]=3$. Linkes Kind $4 \geq$ rechtes Kind 3, wähle linkes Kind



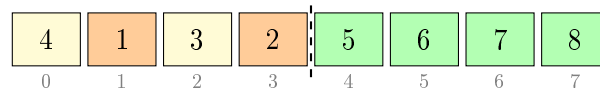
Schritt 54: $a[0]=1 < a[1]=4$ (grösseres Kind), tausche



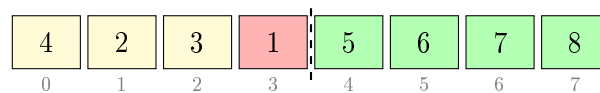
Schritt 55: Nur linkes Kind vorhanden: $a[3]=2$



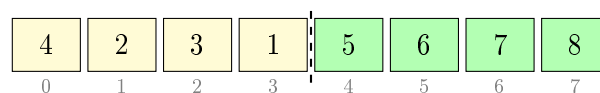
Schritt 56: $a[1]=4 < a[3]=1$ (grösseres Kind), tausche



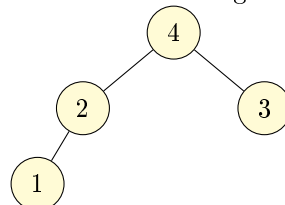
Schritt 57: $a[3]=1$ ist Blatt (keine Kinder), fertig



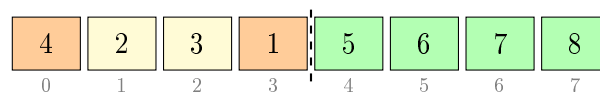
Schritt 58: Heap4: [4, 2, 3, 1] | sortiert: [5, 6, 7, 8]



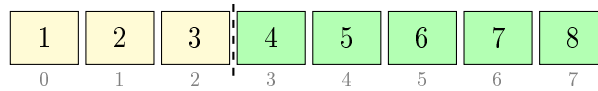
Baumdarstellung:



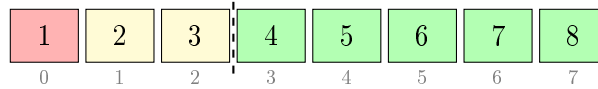
Schritt 59: Tausche Maximum $a[0]=4$ mit $a[3]=1$



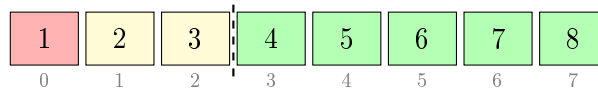
Schritt 60: $a[3]=4$ ist an Endposition, Heapgrösse: 3



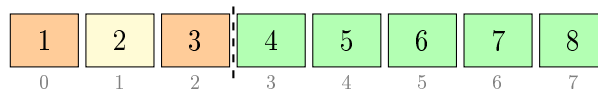
Schritt 61: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=1$



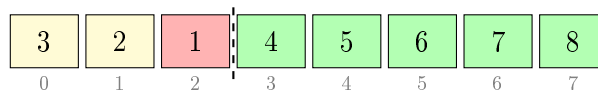
Schritt 62: Kinder: $a[1]=2$ und $a[2]=3$. Rechtes Kind $3 >$ linkes Kind 2 , wähle rechtes Kind



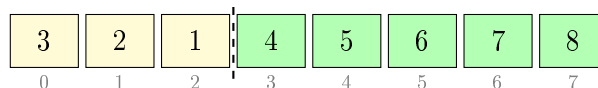
Schritt 63: $a[0]=1 < a[2]=3$ (grösseres Kind), tausche



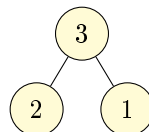
Schritt 64: $a[2]=1$ ist Blatt (keine Kinder), fertig



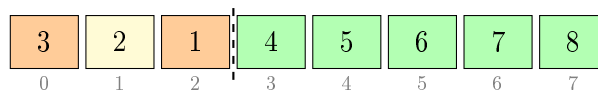
Schritt 65: Heap3: $[3, 2, 1]$ | sortiert: $[4, 5, 6, 7, 8]$



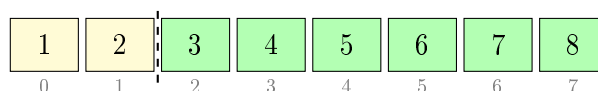
Baumdarstellung:



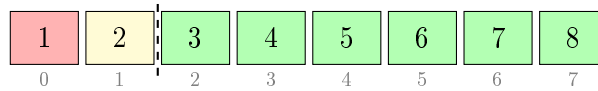
Schritt 66: Tausche Maximum $a[0]=3$ mit $a[2]=1$



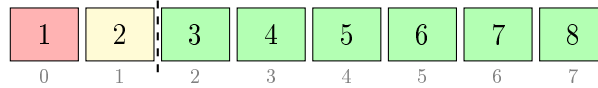
Schritt 67: $a[2]=3$ ist an Endposition, Heapgroesse: 2



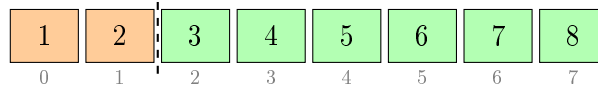
Schritt 68: $\text{siftdown}(a, 0, 2)$: Versickere $a[0]=1$



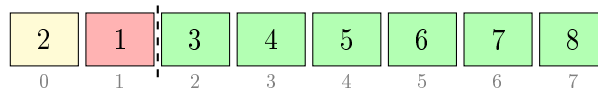
Schritt 69: Nur linkes Kind vorhanden: $a[1]=2$



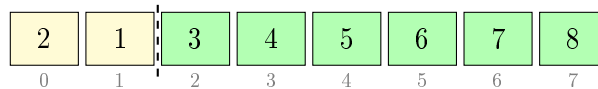
Schritt 70: $a[0]=1 < a[1]=2$ (grösseres Kind), tausche



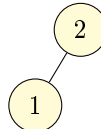
Schritt 71: $a[1]=1$ ist Blatt (keine Kinder), fertig



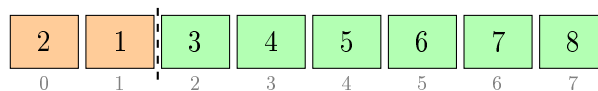
Schritt 72: Heap2: $[2, 1]$ | sortiert: $[3, 4, 5, 6, 7, 8]$



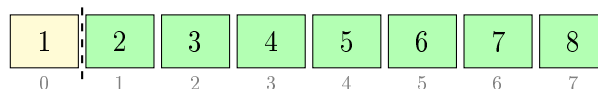
Baumdarstellung:



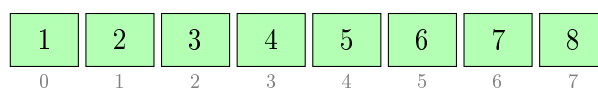
Schritt 73: Tausche Maximum $a[0]=2$ mit $a[1]=1$



Schritt 74: $a[1]=2$ ist an Endposition, Heapgrösse: 1



Schritt 75: Sortierung abgeschlossen: $[1, 2, 3, 4, 5, 6, 7, 8]$



Zusammenfassung der Heap-Zustände:

- **Heap8:** [8, 5, 7, 4, 1, 6, 3, 2]
- **Heap7:** [7, 5, 6, 4, 1, 2, 3] | sortiert: [8]
- **Heap6:** [6, 5, 3, 4, 1, 2] | sortiert: [7, 8]
- **Heap5:** [5, 4, 3, 2, 1] | sortiert: [6, 7, 8]
- **Heap4:** [4, 2, 3, 1] | sortiert: [5, 6, 7, 8]
- **Heap3:** [3, 2, 1] | sortiert: [4, 5, 6, 7, 8]
- **Heap2:** [2, 1] | sortiert: [3, 4, 5, 6, 7, 8]

Antworten:

1. Anfangsheap (Heap8): **[8, 5, 7, 4, 1, 6, 3, 2]**

2. Zwischenheaps:

- Heap7: [7, 5, 6, 4, 1, 2, 3]
- Heap6: [6, 5, 3, 4, 1, 2]
- Heap5: [5, 4, 3, 2, 1]
- Heap4: [4, 2, 3, 1]
- Heap3: [3, 2, 1]
- Heap2: [2, 1]

3. Anzahl Vertauschungen: **22**

Endergebnis: [1, 2, 3, 4, 5, 6, 7, 8]

Aufgabe 6 – Schwierigkeit: ★★★★★

Heapsort mit 10 Elementen (anspruchsvoll)

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

3, 7, 1, 9, 5, 2, 8, 4, 10, 6

Die Liste soll mit dem **Heapsort**-Verfahren sortiert werden.

Fragen:

1. Geben Sie den Anfangsheap (Heap10) an.
2. Geben Sie alle Heaps an, die als Zwischenergebnisse beim Sortieren erzeugt werden (Heap9, ..., Heap3).
3. Wie viele Vertauschungen werden insgesamt durchgefuehrt?

Loesung:

Schritt-fuer-Schritt-Loesung:

Schritt 0: Ausgangszustand

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

=== Phase 1: Anfangsheap herstellen ===

Schritt 1: siftdown(a, 5, 10): Versickere a[5]=2

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

Schritt 2: a[5]=2 ist Blatt (keine Kinder), fertig

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

Schritt 3: siftdown(a, 4, 10): Versickere a[4]=5

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

Schritt 4: Nur linkes Kind vorhanden: a[9]=6

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

Schritt 5: a[4]=5 < a[9]=6 (groesseres Kind), tausche

3	7	1	9	5	2	8	4	10	6
0	1	2	3	4	5	6	7	8	9

Schritt 6: $a[9]=5$ ist Blatt (keine Kinder), fertig

3	7	1	9	6	2	8	4	10	5
0	1	2	3	4	5	6	7	8	9

Schritt 7: $\text{siftdown}(a, 3, 10)$: Versickere $a[3]=9$

3	7	1	9	6	2	8	4	10	5
0	1	2	3	4	5	6	7	8	9

Schritt 8: Kinder: $a[7]=4$ und $a[8]=10$. Rechtes Kind $10 >$ linkes Kind 4, waehe rechtes Kind

3	7	1	9	6	2	8	4	10	5
0	1	2	3	4	5	6	7	8	9

Schritt 9: $a[3]=9 < a[8]=10$ (groesseres Kind), tausche

3	7	1	9	6	2	8	4	10	5
0	1	2	3	4	5	6	7	8	9

Schritt 10: $a[8]=9$ ist Blatt (keine Kinder), fertig

3	7	1	10	6	2	8	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 11: $\text{siftdown}(a, 2, 10)$: Versickere $a[2]=1$

3	7	1	10	6	2	8	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 12: Kinder: $a[5]=2$ und $a[6]=8$. Rechtes Kind $8 >$ linkes Kind 2, waehe rechtes Kind

3	7	1	10	6	2	8	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 13: $a[2]=1 < a[6]=8$ (groesseres Kind), tausche

3	7	1	10	6	2	8	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 14: $a[6]=1$ ist Blatt (keine Kinder), fertig

3	7	8	10	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 15: $\text{siftdown}(a, 1, 10)$: Versickere $a[1]=7$

3	7	8	10	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 16: Kinder: $a[3]=10$ und $a[4]=6$. Linkes Kind $10 \geq$ rechtes Kind 6, waehe linkes Kind

3	7	8	10	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 17: $a[1]=7 < a[3]=10$ (groesseres Kind), tausche

3	7	8	10	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 18: Kinder: $a[7]=4$ und $a[8]=9$. Rechtes Kind $9 >$ linkes Kind 4, waehe rechtes Kind

3	10	8	7	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 19: $a[3]=7 < a[8]=9$ (groesseres Kind), tausche

3	10	8	7	6	2	1	4	9	5
0	1	2	3	4	5	6	7	8	9

Schritt 20: $a[8]=7$ ist Blatt (keine Kinder), fertig

3	10	8	9	6	2	1	4	7	5
0	1	2	3	4	5	6	7	8	9

Schritt 21: $\text{siftdown}(a, 0, 10)$: Versickere $a[0]=3$

3	10	8	9	6	2	1	4	7	5
0	1	2	3	4	5	6	7	8	9

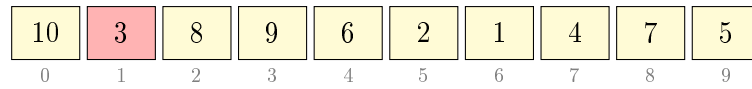
Schritt 22: Kinder: $a[1]=10$ und $a[2]=8$. Linkes Kind $10 \geq$ rechtes Kind 8, waehe linkes Kind

3	10	8	9	6	2	1	4	7	5
0	1	2	3	4	5	6	7	8	9

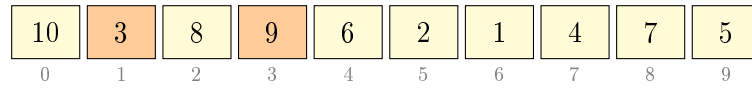
Schritt 23: $a[0]=3 < a[1]=10$ (groesseres Kind), tausche

3	10	8	9	6	2	1	4	7	5
0	1	2	3	4	5	6	7	8	9

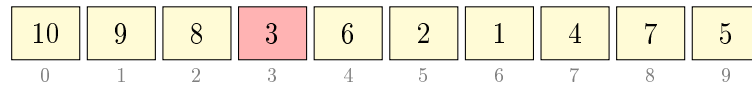
Schritt 24: Kinder: $a[3]=9$ und $a[4]=6$. Linkes Kind $9 \geq$ rechtes Kind 6, waehe linkes Kind



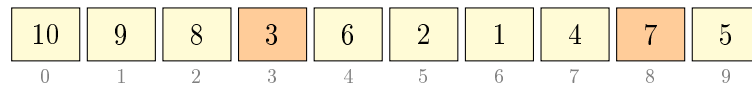
Schritt 25: $a[1]=3 < a[3]=9$ (groesseres Kind), tausche



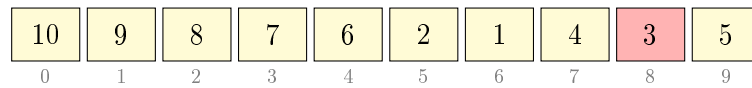
Schritt 26: Kinder: $a[7]=4$ und $a[8]=7$. Rechtes Kind $7 >$ linkes Kind 4 , waehle rechtes Kind



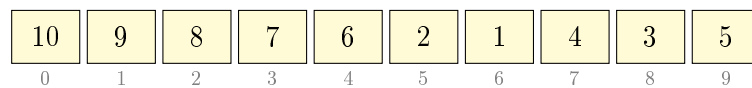
Schritt 27: $a[3]=3 < a[8]=7$ (groesseres Kind), tausche



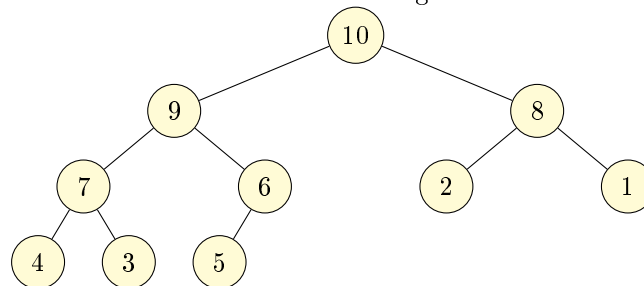
Schritt 28: $a[8]=3$ ist Blatt (keine Kinder), fertig



Schritt 29: Heap10: Anfangsheap hergestellt

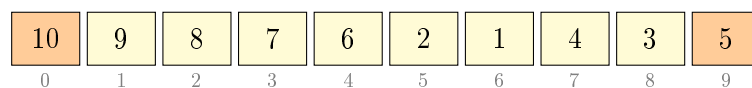


Baumdarstellung:

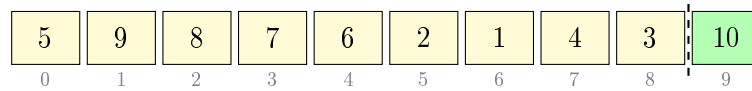


=== Phase 2: Sortieren durch Auswahl ===

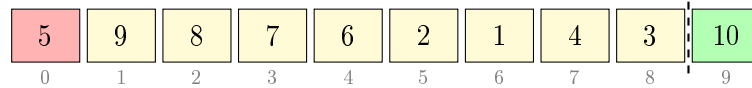
Schritt 30: Tausche Maximum $a[0]=10$ mit $a[9]=5$



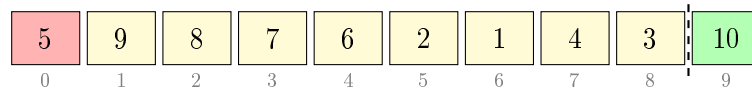
Schritt 31: $a[9]=10$ ist an Endposition, Heapgroesse: 9



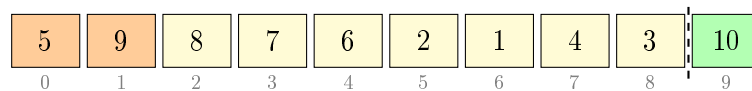
Schritt 32: siftdown(a, 0, 9): Versickere $a[0]=5$



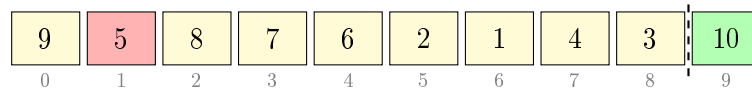
Schritt 33: Kinder: $a[1]=9$ und $a[2]=8$. Linkes Kind $9 \geq$ rechtes Kind 8, waehle linkes Kind



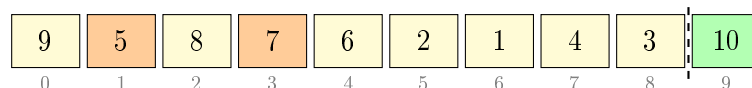
Schritt 34: $a[0]=5 < a[1]=9$ (groesseres Kind), tausche



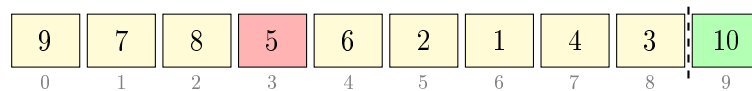
Schritt 35: Kinder: $a[3]=7$ und $a[4]=6$. Linkes Kind $7 \geq$ rechtes Kind 6, waehle linkes Kind



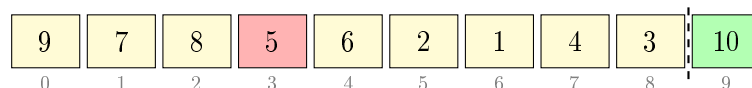
Schritt 36: $a[1]=5 < a[3]=7$ (groesseres Kind), tausche



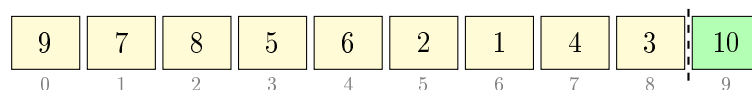
Schritt 37: Kinder: $a[7]=4$ und $a[8]=3$. Linkes Kind $4 \geq$ rechtes Kind 3, waehle linkes Kind

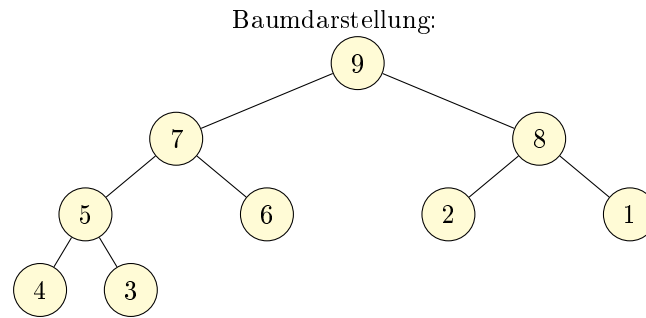


Schritt 38: $a[3]=5 \geq a[7]=4$ (groesseres Kind), Heap-Bedingung erfuehlt

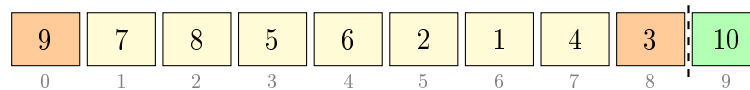


Schritt 39: Heap9: [9, 7, 8, 5, 6, 2, 1, 4, 3] | sortiert: [10]

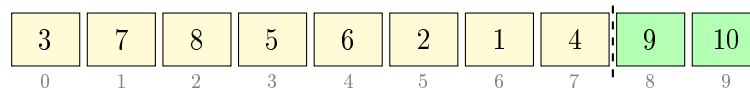




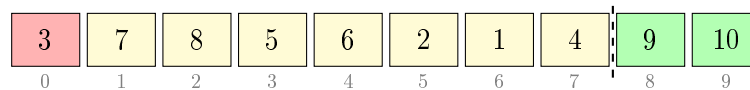
Schritt 40: Tausche Maximum $a[0]=9$ mit $a[8]=3$



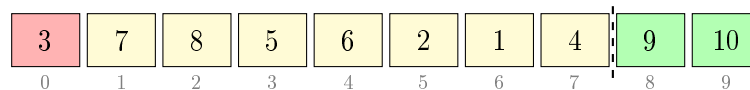
Schritt 41: $a[8]=9$ ist an Endposition, Heapgroesse: 8



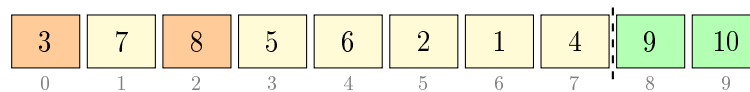
Schritt 42: $\text{siftdown}(a, 0, 8)$: Versickere $a[0]=3$



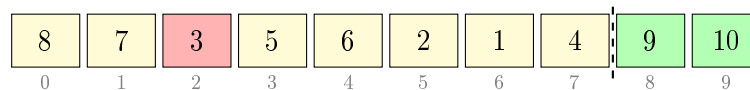
Schritt 43: Kinder: $a[1]=7$ und $a[2]=8$. Rechtes Kind $8 >$ linkes Kind 7 , waehle rechtes Kind



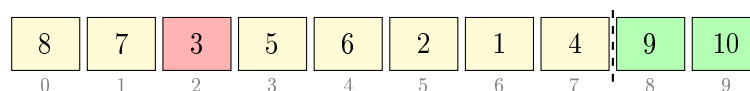
Schritt 44: $a[0]=3 < a[2]=8$ (groesseres Kind), tausche



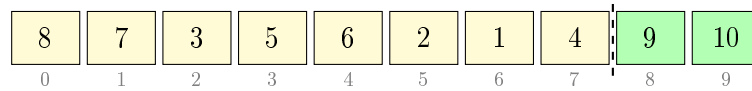
Schritt 45: Kinder: $a[5]=2$ und $a[6]=1$. Linkes Kind $2 \geq$ rechtes Kind 1 , waehle linkes Kind



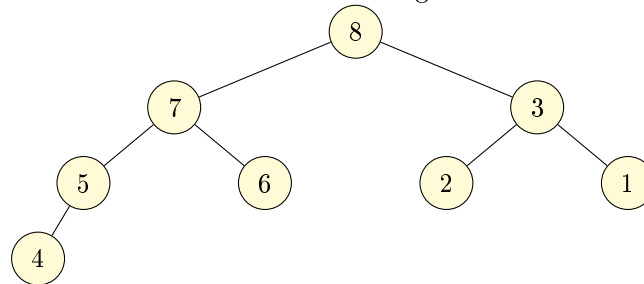
Schritt 46: $a[2]=3 \geq a[5]=2$ (groesseres Kind), Heap-Bedingung erfuehlt



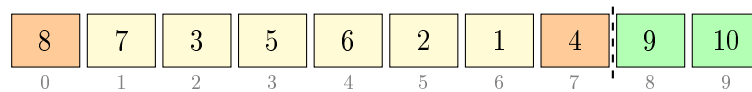
Schritt 47: Heap8: $[8, 7, 3, 5, 6, 2, 1, 4]$ | sortiert: $[9, 10]$



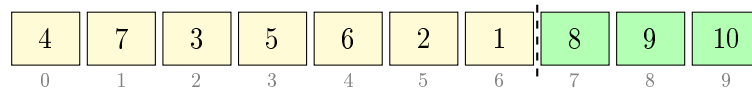
Baumdarstellung:



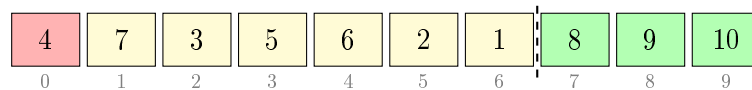
Schritt 48: Tausche Maximum $a[0]=8$ mit $a[7]=4$



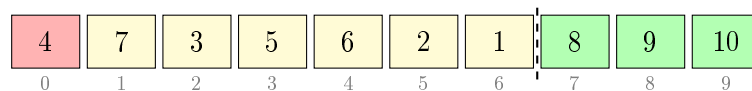
Schritt 49: $a[7]=8$ ist an Endposition, Heapgroesse: 7



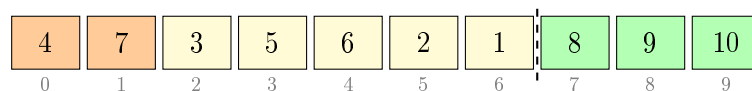
Schritt 50: $\text{siftdown}(a, 0, 7)$: Versickere $a[0]=4$



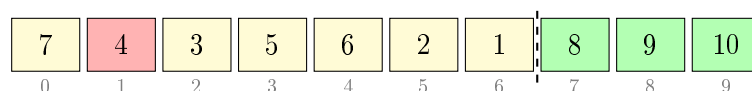
Schritt 51: Kinder: $a[1]=7$ und $a[2]=3$. Linkes Kind $7 \geq$ rechtes Kind 3, wähle linkes Kind



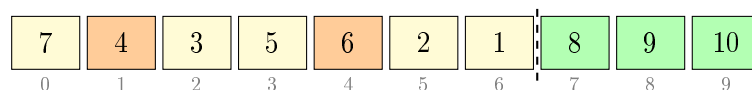
Schritt 52: $a[0]=4 < a[1]=7$ (grösseres Kind), tausche



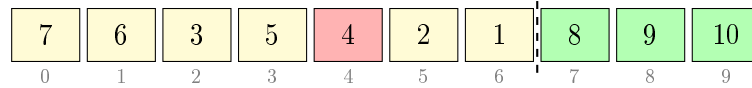
Schritt 53: Kinder: $a[3]=5$ und $a[4]=6$. Rechtes Kind $6 >$ linkes Kind 5, wähle rechtes Kind



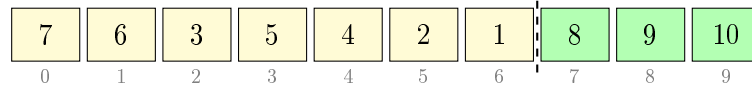
Schritt 54: $a[1]=4 < a[4]=6$ (grösseres Kind), tausche



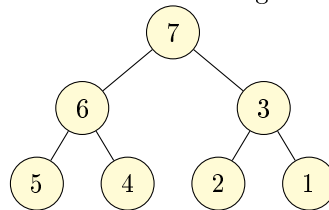
Schritt 55: $a[4]=4$ ist Blatt (keine Kinder), fertig



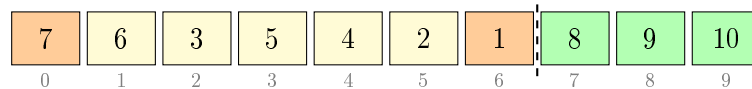
Schritt 56: Heap7: [7, 6, 3, 5, 4, 2, 1] | sortiert: [8, 9, 10]



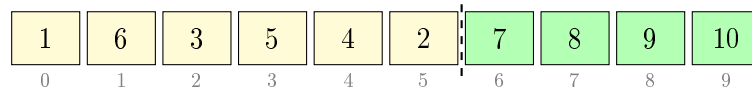
Baumdarstellung:



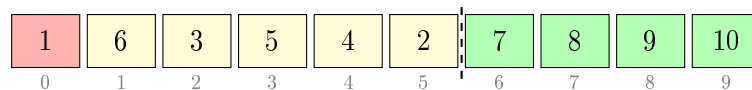
Schritt 57: Tausche Maximum $a[0]=7$ mit $a[6]=1$



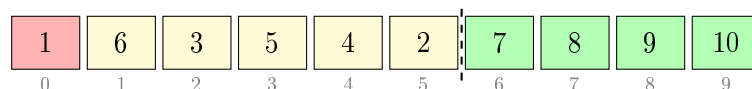
Schritt 58: $a[6]=7$ ist an Endposition, Heapgrösse: 6



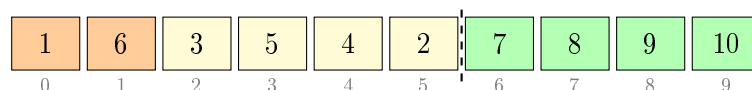
Schritt 59: $\text{siftdown}(a, 0, 6)$: Versickere $a[0]=1$



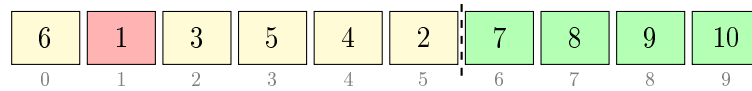
Schritt 60: Kinder: $a[1]=6$ und $a[2]=3$. Linkes Kind $6 \geq$ rechtes Kind 3, wähle linkes Kind



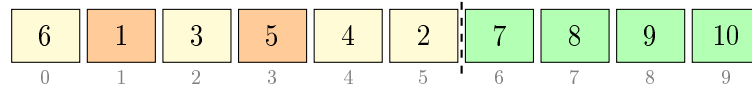
Schritt 61: $a[0]=1 < a[1]=6$ (grösseres Kind), tausche



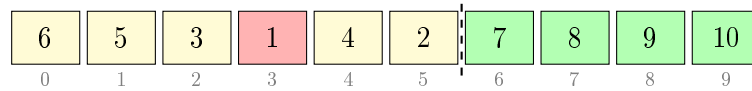
Schritt 62: Kinder: $a[3]=5$ und $a[4]=4$. Linkes Kind $5 \geq$ rechtes Kind 4, wähle linkes Kind



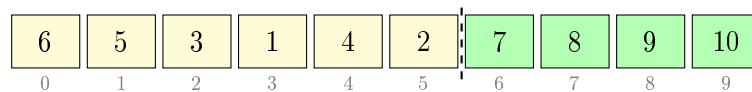
Schritt 63: $a[1]=1 < a[3]=5$ (groesseres Kind), tausche



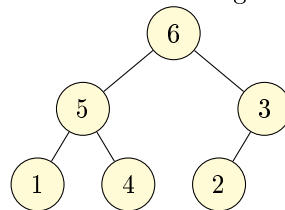
Schritt 64: $a[3]=1$ ist Blatt (keine Kinder), fertig



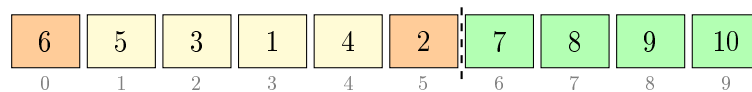
Schritt 65: Heap6: [6, 5, 3, 1, 4, 2] | sortiert: [7, 8, 9, 10]



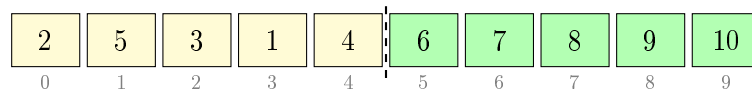
Baumdarstellung:



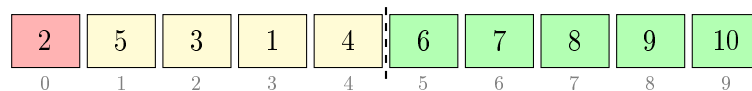
Schritt 66: Tausche Maximum $a[0]=6$ mit $a[5]=2$



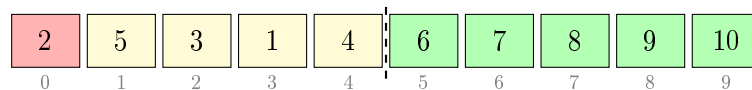
Schritt 67: $a[5]=6$ ist an Endposition, Heapgroesse: 5



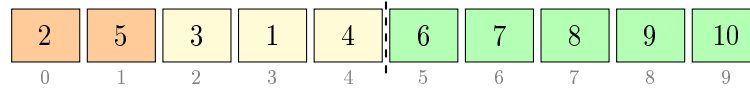
Schritt 68: $\text{siftdown}(a, 0, 5)$: Versickere $a[0]=2$



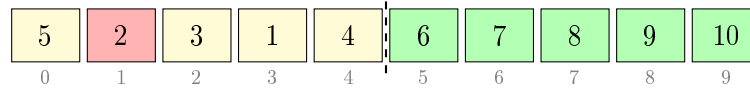
Schritt 69: Kinder: $a[1]=5$ und $a[2]=3$. Linkes Kind $5 \geq$ rechtes Kind 3, waehe linkes Kind



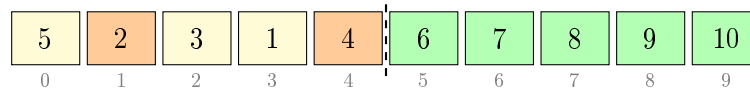
Schritt 70: $a[0]=2 < a[1]=5$ (groesseres Kind), tausche



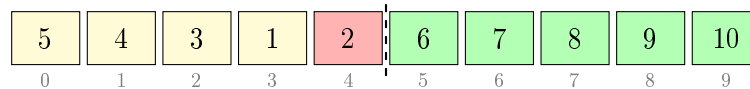
Schritt 71: Kinder: $a[3]=1$ und $a[4]=4$. Rechtes Kind $4 >$ linkes Kind 1 , wachle rechtes Kind



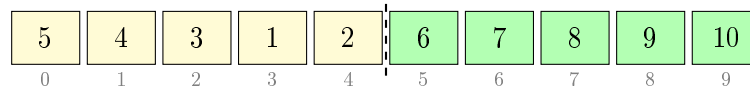
Schritt 72: $a[1]=2 < a[4]=4$ (groesseres Kind), tausche



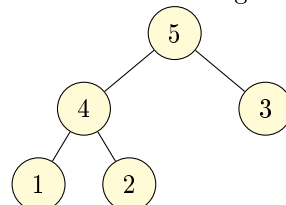
Schritt 73: $a[4]=2$ ist Blatt (keine Kinder), fertig



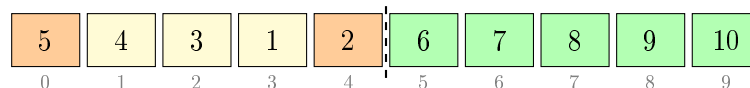
Schritt 74: Heap5: $[5, 4, 3, 1, 2]$ | sortiert: $[6, 7, 8, 9, 10]$



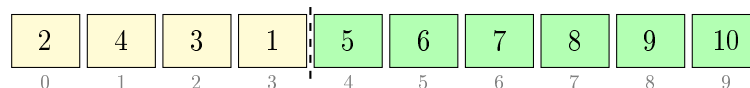
Baumdarstellung:



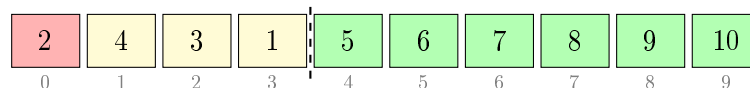
Schritt 75: Tausche Maximum $a[0]=5$ mit $a[4]=2$



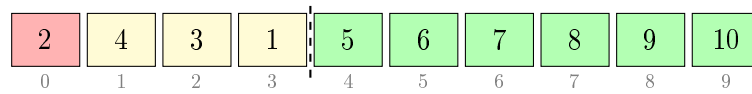
Schritt 76: $a[4]=5$ ist an Endposition, Heapgroesse: 4



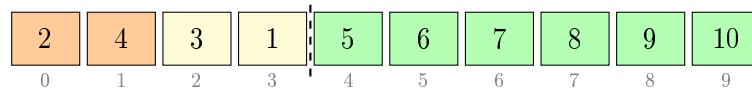
Schritt 77: $\text{siftdown}(a, 0, 4)$: Versickere $a[0]=2$



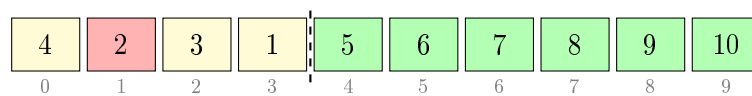
Schritt 78: Kinder: $a[1]=4$ und $a[2]=3$. Linkes Kind $4 \geq$ rechtes Kind 3, waehle linkes Kind



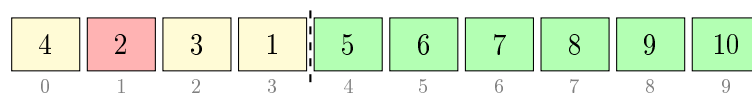
Schritt 79: $a[0]=2 < a[1]=4$ (groesseres Kind), tausche



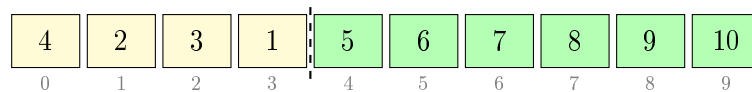
Schritt 80: Nur linkes Kind vorhanden: $a[3]=1$



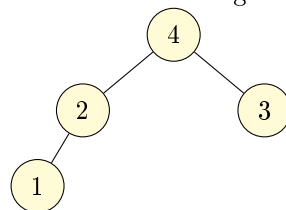
Schritt 81: $a[1]=2 \geq a[3]=1$ (groesseres Kind), Heap-Bedingung erfuehlt



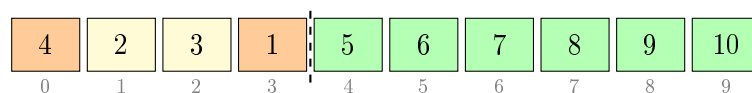
Schritt 82: Heap4: [4, 2, 3, 1] | sortiert: [5, 6, 7, 8, 9, 10]



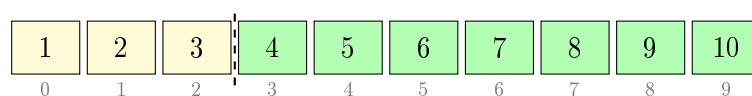
Baumdarstellung:



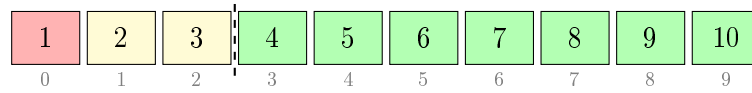
Schritt 83: Tausche Maximum $a[0]=4$ mit $a[3]=1$



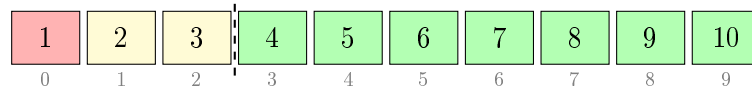
Schritt 84: $a[3]=4$ ist an Endposition, Heapgroesse: 3



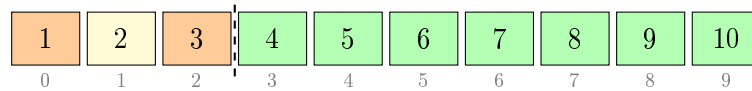
Schritt 85: $\text{siftdown}(a, 0, 3)$: Versickere $a[0]=1$



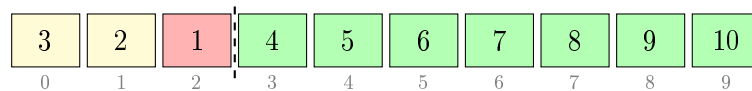
Schritt 86: Kinder: $a[1]=2$ und $a[2]=3$. Rechtes Kind $3 >$ linkes Kind 2, wähle rechtes Kind



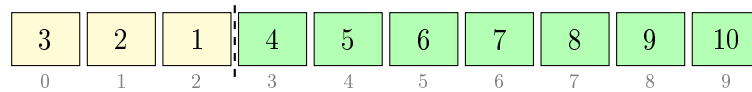
Schritt 87: $a[0]=1 < a[2]=3$ (grösseres Kind), tausche



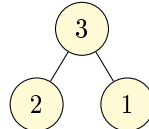
Schritt 88: $a[2]=1$ ist Blatt (keine Kinder), fertig



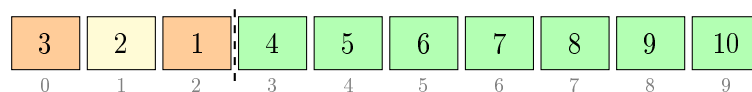
Schritt 89: Heap3: [3, 2, 1] | sortiert: [4, 5, 6, 7, 8, 9, 10]



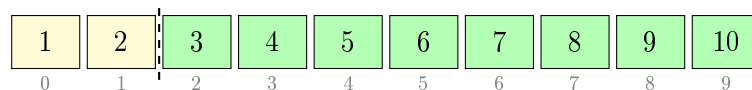
Baumdarstellung:



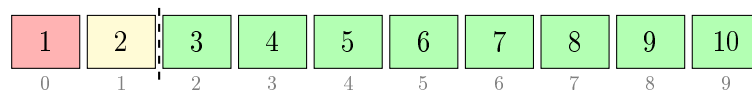
Schritt 90: Tausche Maximum $a[0]=3$ mit $a[2]=1$



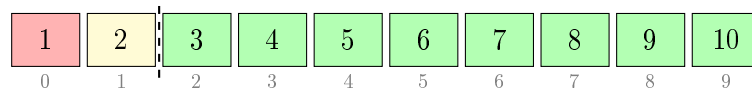
Schritt 91: $a[2]=3$ ist an Endposition, Heapgrösse: 2



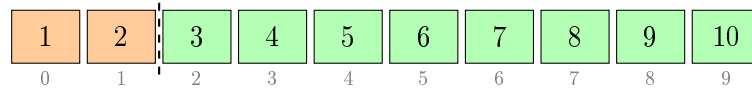
Schritt 92: $\text{sift_down}(a, 0, 2)$: Versickere $a[0]=1$



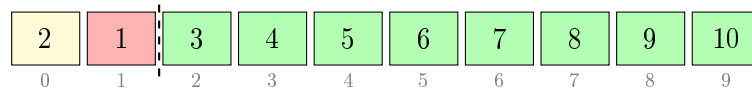
Schritt 93: Nur linkes Kind vorhanden: $a[1]=2$



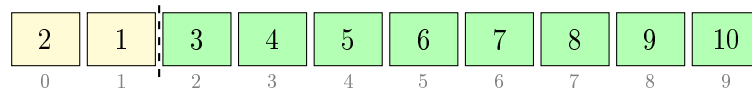
Schritt 94: $a[0]=1 < a[1]=2$ (großeres Kind), tausche



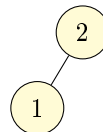
Schritt 95: $a[1]=1$ ist Blatt (keine Kinder), fertig



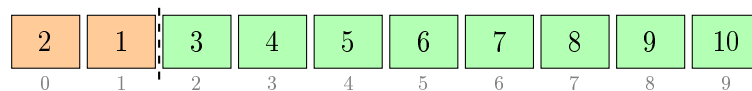
Schritt 96: Heap2: [2, 1] | sortiert: [3, 4, 5, 6, 7, 8, 9, 10]



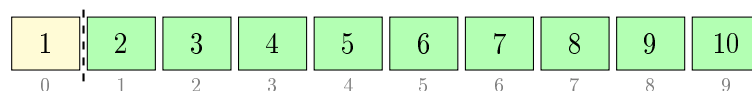
Baumdarstellung:



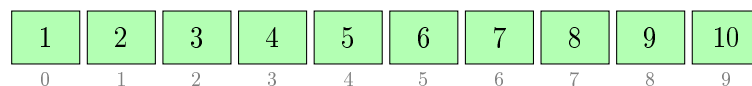
Schritt 97: Tausche Maximum $a[0]=2$ mit $a[1]=1$



Schritt 98: $a[1]=2$ ist an Endposition, Heapgröße: 1



Schritt 99: Sortierung abgeschlossen: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]



Zusammenfassung der Heap-Zustände:

- **Heap10:** [10, 9, 8, 7, 6, 2, 1, 4, 3, 5]
- **Heap9:** [9, 7, 8, 5, 6, 2, 1, 4, 3] | sortiert: [10]
- **Heap8:** [8, 7, 3, 5, 6, 2, 1, 4] | sortiert: [9, 10]

- **Heap7:** [7, 6, 3, 5, 4, 2, 1] | sortiert: [8, 9, 10]
- **Heap6:** [6, 5, 3, 1, 4, 2] | sortiert: [7, 8, 9, 10]
- **Heap5:** [5, 4, 3, 1, 2] | sortiert: [6, 7, 8, 9, 10]
- **Heap4:** [4, 2, 3, 1] | sortiert: [5, 6, 7, 8, 9, 10]
- **Heap3:** [3, 2, 1] | sortiert: [4, 5, 6, 7, 8, 9, 10]
- **Heap2:** [2, 1] | sortiert: [3, 4, 5, 6, 7, 8, 9, 10]

Antworten:

1. Anfangsheap (Heap10): **[10, 9, 8, 7, 6, 2, 1, 4, 3, 5]**

2. Zwischenheaps:

- Heap9: [9, 7, 8, 5, 6, 2, 1, 4, 3]
- Heap8: [8, 7, 3, 5, 6, 2, 1, 4]
- Heap7: [7, 6, 3, 5, 4, 2, 1]
- Heap6: [6, 5, 3, 1, 4, 2]
- Heap5: [5, 4, 3, 1, 2]
- Heap4: [4, 2, 3, 1]
- Heap3: [3, 2, 1]
- Heap2: [2, 1]

3. Anzahl Vertauschungen: **29**

Endergebnis: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
